

Predicting Arabica and Robusta Coffee Futures Prices: An Econometric and Machine Learning Approach with Macroeconomic, Climatic, and Speculative Determinants

BY

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Abstract

This thesis compares six forecasting models, Linear Regression, ARIMAX, Random Forest, XGBoost, Support Vector Machine, and Artificial Neural Network, in predicting monthly ICE Arabica and ICE Futures Europe Robusta coffee futures prices. Arabica is estimated over 211 monthly observations (June 2006–December 2023) and Robusta over 192 (January 2008–December 2023), the periods for which all fourteen predictors are simultaneously available. These predictors span four macroeconomic and financial variables (CPI, WTI, the Federal Funds Rate, DXY), sugar and cocoa futures prices as market-spillover variables, four USDA supply-and-demand fundamentals, temperature and precipitation indices for the leading producer countries, the Oceanic Niño Index, and CFTC net Managed Money positioning. All models are evaluated under an identical walk-forward expanding-window protocol with an 80/20 train-test split, ensuring genuine out-of-sample comparability across econometric and machine learning approaches. ARIMAX, the only model to combine an explicit autoregressive structure with the fourteen exogenous determinants, achieves the strongest out-of-sample performance for both varieties ($R^2 = 0.955$ for Arabica, 0.926 for Robusta), outperforming every machine learning model tested; among the four machine learning models, XGBoost performs best ($R^2 = 0.812$ and 0.797 , respectively). Linear Regression proves highly sensitive to multicollinearity, collapsing to an R^2 of 0.276 for Robusta despite a respectable 0.784 for Arabica. Permutation importance shows that Ending stocks, COT net Managed Money, and WTI matter most for Arabica, while WTI and the Federal Funds Rate matter most for Robusta, a divide not anticipated by the existing literature.

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AI Disclosure

Claude¹ (Anthropic) was used as an artificial intelligence assistant to support the writing, structuring, and tabulation of this thesis, as well as the design and debugging of the data processing pipeline in R.

¹ <https://claude.ai>

Table of Contents

Abstract	i
Acknowledgements	ii
AI Disclosure	iii
1. Introduction	1
2. Literature Review	4
2.1 Coffee Futures Markets as an Object of Forecasting	4
2.2 Financialization of Futures Markets and the Role of Speculative Positioning	6
2.3 Climatic Determinants of Coffee Production and Price Volatility	7
2.4 Forecasting Models Applied to Agricultural Commodity Prices	9
3. Data and Methodology	14
3.1 Sample Period and Data Overview	14
3.2 Dependent Variables	15
3.3 Explanatory Variables	16
3.3.1 Market Spillover Variables	16
3.3.2 Macroeconomic and Financial Variables	16
3.3.3 Supply-and-Demand Fundamentals	17
3.3.4 Speculative Positioning	18
3.3.5 Climate Variables	19
3.4 Correlation Structure of the Predictor Set	23
3.5 Estimation Strategy	24
3.5.1 Model Overview	24
3.5.2 Walk-Forward Validation Protocol	24
3.5.3 Data Preprocessing and Stationarity	25
3.5.4 ARIMAX	26
3.5.5 Linear Regression	27
3.5.6 Random Forest	28
3.5.7 XGBoost	28
3.5.8 Support Vector Machine	29
3.5.9 Artificial Neural Network	30
3.5.10 Model Interpretation: Permutation Importance and Partial Dependence	30
4. Results	33
4.1 Model Performance	33
4.1.1 Adjustment Quality	34
4.1.2 Forecast Trajectories	35
4.2 Model Interpretation	37

4.2.1 <i>Econometric Models</i>	38
4.2.2 <i>Variable Importance and Partial Dependence</i>	39
5. Discussion	43
5.1 Overall Model Performance.....	43
5.2 The Role of Financial and Speculative Variables.....	45
5.3 Arabica vs Robusta: Structural Differences.....	46
6. Conclusion	48
7. References.....	50
8. Appendices.....	55
8.1 Stationarity Testing: ADF and KPSS Procedures.....	55
8.2 MLR and ARIMAX: Coefficient Diagnostics and Robustness.....	55
8.3 Random Forest, XGBoost and SVM: Hyperparameter Sensitivity	59
8.4 Artificial Neural Network: Extensions and Sensitivity Analysis	63
8.5 Variable Importance and Partial Dependence: Full Detail	66
8.6 Robustness Checks.....	75
8.6.1 <i>Disaggregated Climate Variables</i>	75
8.6.2 <i>Variety-Specific Model Specification</i>	76
8.6.3 <i>OECD GDP Growth (Quarterly Specification)</i>	78
8.6.4 <i>Lagged COT Net Managed Money Position</i>	80

1. Introduction

Coffee futures prices respond to a wide array of forces: macroeconomic conditions, climate in producing regions, and the positioning of speculative traders in derivatives markets. Forecasting them accurately, and knowing which of these forces actually drive them, matters to a broad set of actors. Producers and cooperatives carry the price risk directly. Roasters and traders manage procurement costs around it. Policymakers watch it for what it implies about the welfare of coffee-dependent economies. Yet the empirical literature on coffee price forecasting remains comparatively thin. As the next section documents, it has rarely combined a rich set of exogenous predictors with a systematic comparison of econometric and machine learning approaches, and rarer still has it done so on the specific futures contracts that anchor global coffee pricing.

Section 2 reviews this literature in detail and identifies four gaps that motivate the present thesis. Coffee-specific forecasting studies rely almost exclusively on univariate or lightly specified models, with no macroeconomic, climatic, or financial variables built in. Most also work from spot prices or generic commodity indices rather than the ICE Arabica and LIFFE Robusta contracts that actually set the global benchmark. To the best of current knowledge, no study has modelled both varieties within a shared multivariate framework using a common set of predictors. And despite mounting evidence of the financialization of commodity futures markets, CFTC speculative positioning data has yet to be evaluated as a predictor in an out-of-sample coffee price forecasting framework, as opposed to explaining price movements that have already occurred.

These gaps translate into the following four research questions:

RQ1. How accurately can Arabica and Robusta coffee futures prices be forecast out-of-sample when a shared set of macroeconomic, climatic, fundamental, and speculative determinants is incorporated into a multivariate forecasting framework?

RQ2. How do the four machine learning models considered (Random Forest, XGBoost, Support Vector Machine, and Artificial Neural Network) perform, in terms of out-of-sample forecasting accuracy, relative to the two econometric benchmarks (linear regression and ARIMAX), when both are estimated under a common predictor set and an identical walk-forward validation protocol?

RQ3. Which of the macroeconomic, climatic, fundamental, and speculative determinants considered contribute most to forecasting accuracy for each coffee variety, and do Arabica and Robusta differ systematically in the determinants that matter most, reflecting their distinct market structures and producing geographies?

RQ4. Does speculative positioning by Managed Money traders, as captured by the CFTC Disaggregated Commitments of Traders report, contribute meaningfully to the forecasting accuracy and to the explanatory structure of Arabica and Robusta futures prices?

To address these questions, the present thesis compares six forecasting models: two econometric benchmarks, linear regression (MLR) and ARIMAX, and four machine learning algorithms, Random Forest, XGBoost, Support Vector Machine (SVM), and Artificial Neural Network (ANN). All six draw on a common monthly dataset spanning January 2000 to December 2023, built from fourteen macroeconomic, climatic, fundamental, and speculative predictors, and are evaluated using an identical walk-forward validation protocol for both varieties. The complete data processing pipeline, model code, and rendered analysis are made publicly available online.²

The remainder of this thesis is organized as follows. Section 2 reviews the relevant literature on coffee market dynamics, the financialization of commodity futures markets, the climatic determinants of coffee production, and the forecasting models applied to agricultural commodity prices. Section 3 describes the data sources, the construction of the two dependent

² <https://github.com/MohamedHussein2304/coffee-thesis>

variables and fourteen exogenous predictors, and the estimation strategy underlying the six forecasting models. Section 4 presents the out-of-sample forecasting results, including comparative model performance and forecast trajectories, before turning to interpretation through the coefficient-based readings available for the econometric models and the variable importance and partial dependence analysis computed for the machine learning models. Section 5 discusses the results in light of the research questions stated above. Section 6 concludes with a summary of findings, limitations, and directions for future research.

2. Literature Review

2.1 Coffee Futures Markets as an Object of Forecasting

Coffee is one of the most widely traded agricultural commodities in the world, yet forecasting its price remains a distinctly complex undertaking. This complexity reflects the structural heterogeneity of its two principal varieties, treated in the literature as fundamentally distinct markets rather than a single unified coffee market (*Milas et al., 2004*). Arabica futures are traded on the Intercontinental Exchange (ICE) in New York under the Coffee 'C' contract, the world benchmark for Arabica coffee pricing, under which physical delivery of exchange-grade green beans from up to twenty countries of origin is settled (*ICE Futures U.S., n.d.*). Robusta futures are listed on ICE Futures Europe in London, the former LIFFE exchange, where contracts are priced in U.S. dollars per metric tonne and serve as the global benchmark for Robusta coffee pricing (*ICE Futures Europe, n.d.*). Beyond their geographic separation, the two contracts differ in their pricing conventions: the Coffee 'C' contract is quoted in U.S. cents per pound for a lot of 37,500 pounds (*ICE Futures U.S., n.d.*), while the Robusta contract is priced in U.S. dollars per metric tonne for a lot of ten tonnes (*ICE Futures Europe, n.d.*), a distinction that requires unit harmonization prior to any comparative analysis. Together, these markets serve not only as price benchmarks for the global coffee trade but also as instruments through which producers, exporters, roasters, and financial investors manage price risk. This benchmark role reflects an informational function: because futures contracts are standardized and traded on centralized exchanges, they aggregate the dispersed information of market participants more effectively than spot markets, which remain fragmented by quality, location, and delivery terms (*Cheng & Xiong, 2014*).

The economic significance of these markets is difficult to overstate. While coffee is consumed predominantly in high-income countries, its production is almost exclusively concentrated in developing nations, with over fifty countries across Latin America, Africa, and

Asia engaged in its cultivation and export (*Lewin et al., 2004*). This asymmetry generates a trade flow that typically moves from developing to developed regions (*Ge et al., 2025*), exposing producers to substantial price risks that futures markets are designed, at least in part, to mitigate (*Lewin et al., 2004*). Yet these same markets have proven highly volatile. The International Coffee Organization (*ICO, 2020*) documents the considerable price swings of the 2019–2020 period, attributing them to a combination of supply shocks, declining global inventories, and demand disruptions stemming from the COVID-19 pandemic. More recently, the USDA Foreign Agricultural Service (2025a) projected that global ending stocks would decline for a fifth consecutive year to just 20.1 million bags in 2025/26, even as world production was forecast to reach a record 178.8 million bags. This apparent contradiction reflects five consecutive years of consumption growth outpacing available supplies.

What makes these markets particularly challenging to model is not volatility alone, but the diversity of forces driving it. Unlike equity markets, agricultural commodity markets are shaped by a heterogeneous set of determinants spanning the supply side, including production, stocks, and climate, the demand side, encompassing consumption and economic growth, and the financial side, covering exchange rates, energy prices, and speculative positioning (*Maurice & Davis, 2012*). The influence of financial actors has grown substantially in recent decades, as commodity index investment has made prices increasingly sensitive to the risk appetite of institutional investors rather than to supply and demand conditions alone (*Tang & Xiong, 2012*). An additional layer of complexity arises from the interconnectedness of the two coffee markets themselves. *Ge et al. (2025)* document measurable and time-varying cross-market connectedness between ICE Arabica and LIFFE Robusta futures, with a two-way volatility spillover averaging 16 to 19 percent between the two markets, indicating that shocks originating in one propagate to the other.

2.2 Financialization of Futures Markets and the Role of Speculative Positioning

Over the past two decades, financial investors, including commodity index traders and hedge funds, have become substantial participants in commodity futures markets (*Cheng & Xiong, 2014; Irwin & Sanders, 2011*). This process, commonly referred to as financialization, has generated a broader debate: rather than resolving it directly, Cheng and Xiong (2014) argue that both the view that financialization caused a price bubble and the view that speculation is inconsequential are too simplistic, and that the more productive question concerns how financial investors have altered the underlying functioning of these markets. Its empirical consequences for the structure of commodity price co-movement are nonetheless well documented and carry direct implications for forecasting.

Tang and Xiong (2012) document a sharp increase in the correlation between non-energy commodities and crude oil prices beginning around 2004, coinciding with the large-scale entry of commodity index investors into futures markets, interpreting this as evidence that index investment introduced a new source of systematic risk linking otherwise heterogeneous commodities. For coffee futures, this implies that macro-financial variables such as oil prices and speculative positioning may carry predictive content beyond what supply and demand fundamentals alone provide.

Complementary evidence at the level of soft commodities reinforces this picture. Maurice and Davis (2012), whose analysis of market efficiency was discussed in the preceding section, also examine the long-run causal structure of price determination across coffee and cocoa, finding that oil price movements exert a causal influence on both Arabica and Robusta prices with transmission lags of approximately four years. Their analysis further suggests that growing speculative activity contributed to the price surges of the late 2000s, pushing prices above levels that fundamentals alone would justify, a result consistent with the financialization narrative of Tang and Xiong (2012) and one that strengthens the case for treating speculative

positioning as a distinct predictor rather than subsuming it within broader macro-financial controls. Ederer et al. (2016), using a different multivariate time series framework than the one applied in this thesis, find that net long positions of Managed Money account for approximately 50% of the variation in Arabica coffee futures prices between 2006 and 2012, the largest share among the five commodities for which they find a statistically significant effect.

Taken together, this body of evidence indicates that speculative positioning is an empirically supported channel through which financialization influences coffee futures prices, rather than a peripheral consideration. The net positioning of managed money traders, derived from the Commodity Futures Trading Commission's Disaggregated Commitments of Traders reports, is therefore included as a predictor in the present thesis on grounds that are both theoretically grounded and empirically supported across multiple methodological frameworks.

2.3 Climatic Determinants of Coffee Production and Price Volatility

Coffee cultivation is among the most climatically sensitive of all major agricultural commodities, and this sensitivity has direct consequences for price dynamics on futures markets. The two principal varieties occupy distinct agronomic niches: Arabica is concentrated in the higher-altitude zones of South America, while Robusta is predominantly grown in the lower-altitude regions of Southeast Asia and Africa, a divide reflected in the variety-specific country groupings used for the climate indices (Section 3.3.5). This geographic divergence means that climate shocks do not affect the two markets symmetrically, and that any forecasting framework treating them as interchangeable would misrepresent the structure of supply-side risk to which each is exposed.

The degree to which temperature constrains coffee production is well documented at the agronomic level. Läderach et al. (2017) show that even modest temperature increases are sufficient to compromise the viability of entire production zones, projecting that areas between 500 and 800 meters above sea level in Nicaragua will lose their agronomic suitability for

Arabica by 2050 as growing conditions shift upward along the altitudinal gradient. While this projection operates on a longer horizon than the forecasting exercise of the present thesis, it establishes that temperature is a first-order constraint on coffee production at the agronomic level, which motivates its inclusion as a predictor alongside the shorter-term variation captured in the monthly series used here.

The El Niño–Southern Oscillation phenomenon has received the most sustained empirical attention as a systematic driver of coffee supply shocks. ENSO describes the interaction between ocean and atmosphere in the tropical Pacific, producing two alternating phases: El Niño, characterized by unusually warm sea surface temperatures in the central and eastern Pacific, and La Niña, associated with colder-than-normal conditions and intensified trade winds (*Ubilava, 2012*). These phases recur every two to seven years and generate geographically differentiated weather anomalies with direct consequences for coffee-growing regions. Because Arabica and Robusta are cultivated in climatically distinct geographies, ENSO shocks affect the two markets through structurally asymmetric supply channels. El Niño tends to create favorable conditions for Arabica production in Central America while simultaneously depressing Robusta output in Southeast Asia, generating opposing price pressures across the two futures markets (*Ubilava, 2012*). La Niña produces the reverse pattern, though its effects on coffee prices tend to be more persistent due to the asymmetric dynamics of the ENSO cycle itself (*Ubilava, 2012*). Overall, Ubilava (2012) finds that ENSO shocks exert a statistically significant influence on world coffee price dynamics, with effects detectable for up to one year following the initial shock.

Bastianin et al. (2018) offer an important qualification to this picture. While confirming that ENSO anomalies exert a statistically significant influence on Colombian Arabica production, exports, and price, they find that the overall explanatory power of ENSO remains modest in variance decomposition terms: in the short run, ENSO shocks account for only 2.2%

of the variation in Colombian coffee production and 0.2% of the variability in its real price, rising to 8.3% and 5.8% respectively in the long run. By contrast, shocks arising in the world coffee market explain 97% of the variation in the real price of Colombian coffee on impact. This result does not undermine the case for including climate variables, but it does clarify their role: rather than standalone predictors capable of driving forecast accuracy on their own, they are best understood as one component of a broader multivariate framework in which macroeconomic and financial forces carry the greater share of explanatory weight.

Taken together, the evidence reviewed in this section points to a consistent conclusion: climate variables, and ENSO indices in particular, carry genuine but bounded explanatory content for coffee futures prices. Their inclusion alongside macroeconomic and financial predictors is empirically warranted, but their relative contribution to forecast accuracy is likely to be modest compared to demand-side and global market forces. This bounded role is precisely what motivates their integration into a broader multivariate framework rather than their treatment as primary drivers, a distinction that shapes both the variable selection and the interpretive expectations of the present thesis.

2.4 Forecasting Models Applied to Agricultural Commodity Prices

Two broad families of models are commonly applied to agricultural commodity price forecasting: time series econometric models such as ARIMA and its multivariate extension ARIMAX, valued for their parsimony and interpretability, and machine learning methods, increasingly used as an alternative for capturing non-linear dynamics. Understanding where each family succeeds and where it fails is a necessary starting point for motivating the comparative framework adopted in this thesis.

Applied to coffee price series, the limitations of purely autoregressive specifications become apparent early. Maurice and Davis (2012), whose account of the heterogeneous supply, demand, and financial determinants of coffee prices already informs the discussion of market

complexity in Section 2.1, provide one of the most comprehensive empirical analyses of price volatility across coffee and related soft commodities, finding that coffee futures markets are broadly efficient in the weak-form sense: if past prices alone contain no exploitable predictive content, meaningful forecasts require the integration of fundamental and financial determinants alongside the price series itself.

Ahumada and Cornejo (2016) forecast quarterly prices of corn, soybeans, and wheat using a multivariate framework estimated on USDA production and inventory data, GDP growth for China, India, and the OECD, the US real exchange rate, and monetary aggregates. They find that models incorporating price interdependence and macroeconomic fundamentals generally outperform single-equation specifications and beat the random walk over a large part of the 2008–2014 out-of-sample period.

Milas et al. (2004) provide an early benchmark, applying linear and non-linear time series models to the spot prices of four coffee types using quarterly data spanning 1962 to 2001. Their out-of-sample evaluation reveals that non-linear specifications improve forecast accuracy over the random walk primarily for Colombian Milds (a washed Arabica origin), while neither class of model consistently outperforms the random walk for the remaining varieties, a result the authors attribute partly to the absence of detectable non-linearity in the forecast period. Their models rely exclusively on the price history of the four series, incorporating no macroeconomic, climatic, or financial determinants. This limitation is illustrated by the 2019–2020 price swings documented in Section 2.1, where the ICO (2020) attributes the disruption to a combination of supply shocks, declining global inventories, and pandemic-driven demand shifts, none of which is represented in a model constrained to its own lagged values.

Deina et al. (2022) show that this limitation persists even in more recent work. Applying a comparative forecasting framework to monthly Arabica and Robusta price series,

benchmarking linear time series models against neural network specifications across horizons of one to twelve steps ahead, they find that non-linear methods outperform linear benchmarks on all standard error metrics, consistently across varieties and horizons. Yet, like Milas et al., their models rely exclusively on price history, without macroeconomic, climatic, or financial determinants.

Evidence beyond coffee presents a more mixed picture. Kohzadi et al. (1996) demonstrated, on monthly cash prices of wheat and live cattle over 1950–1990, that a feedforward neural network with a single hidden layer outperformed an ARIMA benchmark on all standard error metrics while also capturing a statistically significant number of price turning points that the linear model failed to detect. Like Kohzadi et al., the present thesis relies on walk-forward validation, evaluating each model sequentially out of sample, applied at the same monthly frequency to Arabica futures prices from 2006 to 2023 and Robusta futures prices from 2008 to 2023. However, Masini et al. (2023), in a comprehensive survey of supervised machine learning methods for time-series forecasting, find that nonlinear methods do not systematically outperform linear ones: in their own empirical illustration, a penalized linear regression is the best-performing model and random forest underperforms.

Oktoviany et al. (2021) develop a machine learning framework for agricultural commodity futures that incorporates meteorological variables and USDA supply-and-demand fundamentals as predictors. Although their application focuses on maize rather than coffee, the logic of integrating weather, stocks, and production into a machine learning architecture is directly transposable to the present setting, and their use of Refinitiv as the primary data source provides bibliographic precedent for the same choice made in this thesis.

Fang et al. (2020), decomposing six categories of Chinese agricultural futures prices into fast-moving and slow-moving components and testing SVM, neural network, and ARIMA specifications on each, find that SVM and neural networks dominate on the volatile, high-

frequency component while no method holds a systematic advantage on the underlying trend, a result suggesting that the relative performance of ML models over econometric benchmarks depends on the structure of price variation being modelled.

Nayak et al. (2024) provide a particularly close methodological parallel, evaluating ARIMAX and MLR alongside ANN, SVR, Random Forest, and XGBoost, together with several deep learning architectures outside the scope of the present thesis, for the prediction of vegetable prices in India using weather variables as exogenous inputs. Their results reveal a more nuanced hierarchy than a simple split between linear and non-linear methods: the deep learning architectures introduced in the same study clearly dominate, but among the remaining six models, nearly identical to the model set used in the present thesis, the linear econometric benchmarks, ARIMAX and MLR, are frequently competitive with or superior to Random Forest and Support Vector Regression, while the artificial neural network posts the weakest results throughout.

Taken together, the literature surveyed in this section points to four interconnected gaps that the present thesis aims to address. Coffee-specific forecasting studies have relied almost exclusively on univariate or lightly specified models that incorporate no macroeconomic, climatic, or financial determinants, leaving a substantial share of price variance unexplained. Existing studies operate predominantly on spot price series or generic commodity indices rather than on the ICE Arabica and LIFFE Robusta futures contracts that serve as the global benchmarks for coffee pricing. No study has, to the best of current knowledge, modelled both varieties within a shared multivariate forecasting framework incorporating a common set of exogenous predictors, a gap made more consequential by the cross-market connectedness between Arabica and Robusta documented in Section 2.1, and one that separates the present thesis from studies such as Deina et al. (2022), which analyse both varieties but without exogenous determinants. Finally, despite empirical evidence that financialization constitutes a

structurally significant influence on coffee futures markets, speculative positioning data derived from the CFTC Disaggregated Commitments of Traders reports has not been evaluated as a predictor in an out-of-sample coffee price forecasting framework, as opposed to explaining price movements that have already occurred. The present thesis addresses each of these gaps by applying a structured comparison of econometric and machine learning models to monthly ICE Arabica and LIFFE Robusta futures prices drawn from a dataset spanning 2000–2023, under a common set of macroeconomic, climatic, fundamental, and speculative predictors.

3. Data and Methodology

3.1 Sample Period and Data Overview

The empirical analysis covers the period from January 2000 to December 2023, yielding 288 monthly observations in the assembled monthly dataset. The choice of this starting point reflects three converging considerations. First, the suspension of the International Coffee Organization's export quota system in 1989, and its definitive abandonment following the failure of successive revival attempts over the subsequent four years, produced a decade of institutional transition that fundamentally restructured the price formation environment prevailing over the analysis period (*Lewin et al., 2004*). Price data prior to 2000 reflect a structurally distinct market environment whose inclusion would introduce regime-change discontinuities that complicate time series modelling without adding comparably informative variation. Second, the ERA5 reanalysis dataset used for climate variables reaches its highest spatial and temporal reliability from 2000 onward, corresponding to the transition to the modern satellite era documented by (*Hersbach et al., 2020*). Third, the CFTC Disaggregated Commitments of Traders report provides historical data only from June 2006 onward, and Robusta futures prices from Refinitiv are unavailable before January 2008. The analysis period ends in December 2023, the last month for which all data sources were simultaneously available at the time of data collection.

The dataset integrates eight distinct data sources spanning six variable categories: dependent variables, market spillover variables, macroeconomic and financial variables, supply-and-demand fundamentals, speculative positioning variables, and climate variables. Two structural data gaps deserve explicit acknowledgement. Robusta futures prices are available from Refinitiv only from January 2008 onward, so all Robusta model specifications are estimated on the 2008–2023 sub-period. The Managed Money net position is missing for the 77 months preceding June 2006. Since this variable enters all six model specifications as

one of the fourteen exogenous predictors, this constraint also applies to Arabica, whose estimation sample accordingly begins in June 2006 rather than January 2000, yielding 211 monthly observations against Robusta's 192. No imputation is applied to either series; missing values are handled in the modelling phase by restricting each specification to the sub-period for which all required variables are simultaneously available. The two final datasets used throughout the modelling phase are constructed by joining all variable series onto a complete monthly date spine covering January 2000 to December 2023, yielding a base monthly dataset of 288 observations across 16 analytical variables before the sample restrictions described above are applied.

3.2 Dependent Variables

The two dependent variables are the monthly average closing prices of Arabica coffee futures traded on ICE Futures U.S. (contract symbol KC) and Robusta coffee futures traded on ICE Futures Europe, the former London International Financial Futures Exchange (contract symbol RC). Both series are extracted from Refinitiv via the CEDIF data facility at the University of Lausanne (*Refinitiv, 2026*), which constitutes the primary source for price data in this study. Arabica and Robusta are treated as two distinct dependent variables and modelled separately throughout the analysis, reflecting their different market structures, consumer bases and price determinants. A combined coffee price index is not constructed, as aggregation would obscure the variety-specific dynamics that this study aims to capture.

Daily closing prices are aggregated to monthly frequency by computing the arithmetic mean of all trading days within each calendar month.

Table 3.1: Descriptive statistics — dependent variables

Variable	Unit	N	Mean	SD	Min	Max
Arabica price	USD/lb	211	1.51	0.42	0.93	2.84
Robusta price	USD/tonne	192	1,837	349	1,190	2,734

Arabica is estimated over 211 monthly observations (June 2006–December 2023) and Robusta over 192 (January 2008–December 2023), reflecting the data availability constraints documented in Section 3.1.

3.3 Explanatory Variables

3.3.1 Market Spillover Variables

Sugar and cocoa futures prices are included as potential market spillover variables. Coffee and cocoa are both tropical commodities produced predominantly in developing countries and traded on international futures markets (*Maurice & Davis, 2012*). The expansion of sugarcane cultivation in Brazil has occurred partly through the substitution of coffee and other traditional crops (*Gilio & Moraes, 2016*), suggesting a potential supply-side linkage between the two markets that may generate contemporaneous price co-movement. The Sugar No.11 contract traded on ICE Futures U.S. and the ICE New York cocoa contract are used, both sourced from Refinitiv via CEDIF (*Refinitiv, 2026*) and aggregated to monthly frequency.

3.3.2 Macroeconomic and Financial Variables

Four macroeconomic variables are included to capture the broader financial and monetary environment that influences commodity prices. The WTI crude oil price, sourced from the Federal Reserve Economic Data (FRED) database (*Federal Reserve Bank of St. Louis, 2026a*), captures the energy cost dimension of coffee production and transportation (*Ederer et al., 2016*), as well as the broad commodity price cycle that tends to co-move across raw materials. The US Dollar Index (DXY), sourced from Investing.com (*Investing.com, 2026*), measures the value of the US dollar against a basket of major currencies; since coffee futures are denominated in US dollars, a stronger dollar tends to depress prices for non-dollar buyers, reducing global demand. The Federal Funds Rate, also sourced from FRED (*Federal Reserve*

Bank of St. Louis, 2026b), proxies the monetary policy stance of the Federal Reserve and influences global capital flows, risk appetite and commodity price dynamics. The US Consumer Price Index (CPI), sourced from FRED (*Federal Reserve Bank of St. Louis, 2026c*), captures the general inflation environment. All four variables are available at monthly frequency.

OECD quarterly real GDP growth, sourced from the OECD Data Explorer (*OECD, 2026*), is used in a robustness check reported in the Appendix, where the frequency conversion this requires is detailed.

3.3.3 Supply-and-Demand Fundamentals

Supply-and-demand fundamentals are sourced from the USDA Foreign Agricultural Service Production, Supply and Distribution (PSD) Online database (*USDA Foreign Agricultural Service, 2025b*), which publishes annual crop-year estimates for the United States and key producing and consuming countries across all major agricultural commodities. Four variables are retained at the world level: total coffee production, total domestic consumption, total exports, and total ending stocks, each constructed by summing across all countries for the relevant crop year.

Since USDA data are published at annual crop-year frequency, they are expanded to monthly frequency by repeating each year's value across the 12 calendar months of that year. This approach assumes that market participants incorporate annual supply-demand estimates into their pricing decisions throughout the entire crop year, consistent with the premise that fundamental information is continuously reflected in futures prices.

In addition to global aggregates, variety-specific fundamentals are constructed for the robustness checks described in the Appendix. For each of the four supply-demand variables, country-level values are multiplied by empirically computed variety split ratios and then

summed across the countries assigned to the Arabica and Robusta groups respectively, yielding fundamentals that reflect conditions in the actual producing zones for each variety rather than global totals.

3.3.4 Speculative Positioning

The Commitments of Traders (COT) report is a weekly publication of the Commodity Futures Trading Commission (*CFTC, 2026*), the US regulatory authority for futures markets. This study uses the Disaggregated Futures-Only report, which divides market participants into four categories: Producer/Merchant/Processor/User, Swap Dealers, Managed Money, and Other Reportables (*CFTC, n.d.*). The Managed Money category encompasses registered commodity trading advisors, commodity pool operators and unregistered funds identified by the CFTC, engaged in managing and conducting organized futures trading on behalf of clients (*CFTC, n.d.*). This classification is not perfectly clean: money managers pursue a mix of technical and fundamentals-based strategies, and even commercial traders sometimes engage in speculative positioning alongside hedging, so the category is best read as an approximation of active speculative activity rather than a pure measure of it (*Ederer et al., 2016*).

$$COT_{netMM}(t) = \left(\frac{1}{N_t}\right) \cdot \sum [LongMM(k) - ShortMM(k)], k \in t$$

where $LongMM(k)$ and $ShortMM(k)$ denote the number of long and short Coffee C contracts held by the Managed Money category at weekly reporting date k , and N_t is the number of weekly reports published within calendar month t .

The variable constructed is the net position of the Managed Money category on the Coffee C contract, computed as the difference between the number of long and short contracts held by this category at each weekly reporting date. A positive net position indicates that speculative funds hold more long contracts than short, implying a net expectation of rising

Predicting Arabica and Robusta Coffee Futures Prices

coffee prices; a negative net position signals a net bearish stance. Weekly observations are aggregated to monthly frequency by computing the arithmetic mean of all reports published within each calendar month. The series is available from June 2006 onward. This unnormalized, contract-based formulation matches that of Ederer et al. (2016), who construct the same measure for Managed Money on the ICE coffee contract using the same CFTC report.

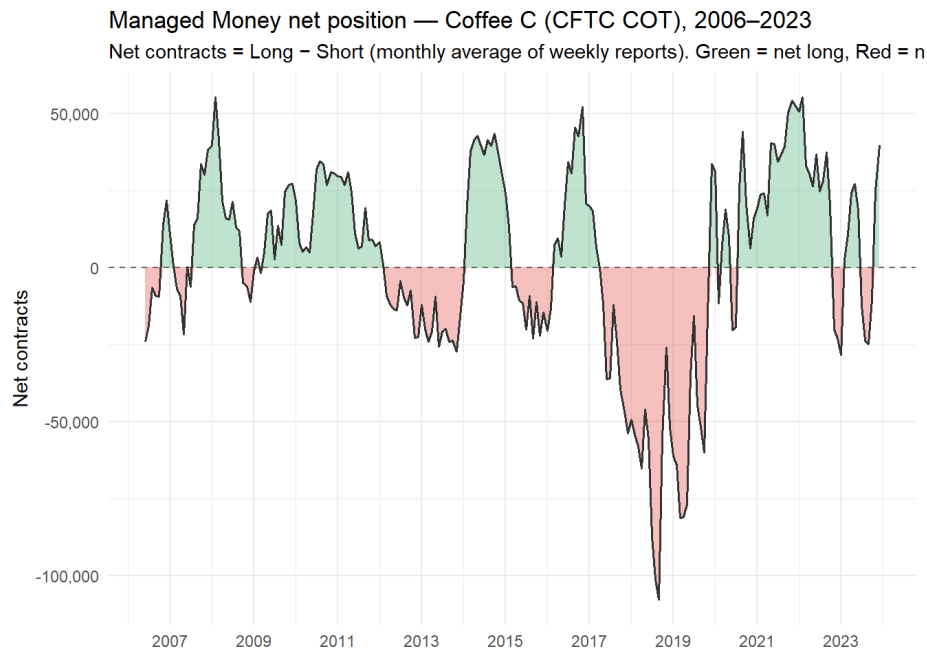


Figure 3.1: Managed Money net position — Coffee C (CFTC COT), 2006–2023

Figure 3.1 plots the monthly COT net Managed Money position for the Coffee C contract over the full 2006–2023 estimation period. The series alternates between clearly positive and negative stretches rather than trending steadily in one direction, a pattern Ederer et al. (2016) also document for money managers and attribute to their more active, shorter-term trading strategies, in contrast to the persistently positive net positions typically held by swap dealers.

3.3.5 Climate Variables

Selection of producer countries.

Temperature and precipitation data are constructed as production-weighted indices aggregated across the 14 countries that individually account for more than 1% of world coffee production on average over the 2000–2024 period. This threshold is applied directly to the USDA PSD data used for the supply fundamentals variables, ensuring full consistency between the production weights and the underlying data source. The 14 countries retained, namely Brazil, Vietnam, Colombia, Indonesia, Ethiopia, India, Honduras, Mexico, Uganda, Guatemala, Peru, Côte d'Ivoire, Nicaragua and Costa Rica, collectively account for approximately 90.8% of world coffee production over the analysis period.

Data source and spatial resolution.

Temperature and precipitation data are sourced from the World Bank Climate Change Knowledge Portal (*World Bank, 2026*), which provides ERA5 reanalysis data at a spatial resolution of 0.25 degrees, offering detailed spatial coverage across all 14 producer countries over the full 2000–2023 period. ERA5 is originally produced by the Copernicus Climate Change Service at the European Centre for Medium-Range Weather Forecasts (*Hersbach et al., 2020*). Data are extracted at Administrative Division Level 1, that is, at the province or state level within each country. For each country, a national monthly average is then computed by taking the simple arithmetic mean across all ADM1 regions, yielding a single national climate series per country and per month.

Index construction.

The production-weighted global index is then defined as:

$$Index(t) = \sum w_i \cdot X_i(t), \quad i = 1, \dots, 14$$

where $X_i(t)$ is the national average value of the climate variable for country i in month t , and w_i is the normalised production weight of country i , with $\sum w_i = 1$. Production weights are held constant at their 2000–2024 averages rather than varying year by year. This fixed-

weight approach avoids a simultaneity problem: if a climate shock in year t reduces production in a given country, its production share would be lower precisely in the year when the shock is most severe, causing time-varying weights to attenuate the very signal the index is designed to capture.

In addition to the global indices described above, variety-specific climate indices are constructed for the robustness check described in the Appendix. The same weighted-average formula is applied separately to the twelve Arabica-producing and six Robusta-producing countries, using production weights renormalised within each variety group according to the same empirically computed variety split ratios applied to the supply-demand fundamentals in Section 3.3.3.

Oceanic Niño Index.

The Oceanic Niño Index (ONI), published by the National Oceanic and Atmospheric Administration (*NOAA*, 2026), measures the three-month running average of sea surface temperature anomalies in the Niño 3.4 region of the central equatorial Pacific Ocean, expressed as a departure from a 30-year climatological baseline. Unlike the national temperature and precipitation indices described above, the ONI does not directly measure conditions in coffee-growing zones but captures the state of the ENSO cycle, whose two phases generate geographically differentiated weather anomalies across producing regions with well-documented consequences for supply. Its inclusion therefore reflects a distinct mechanism: beyond its effect on realised production, ENSO conditions may influence futures prices through anticipatory market behaviour, as traders adjust their positioning in response to developing climate signals before supply consequences fully materialise. This anticipatory channel, along with the expected negative relationship between ONI and Colombian Arabica prices, is documented directly by Bastianin et al. (2018), who find that a positive ENSO shock (El Niño)

Predicting Arabica and Robusta Coffee Futures Prices

raises Colombian coffee production and exports with a lag of twelve to sixteen months, thereby lowering the real price of Colombian coffee, while a negative shock (La Niña) produces the opposite pattern. The anomaly column is retained rather than the absolute sea surface temperature, as it captures deviation from normal conditions and is the operationally relevant measure for assessing ENSO strength. The NOAA file reports values for overlapping three-month seasons (DJF, JFM, FMA and so on); each season is assigned to its central month, so that the ONI series aligns with the monthly frequency of all other variables.

Table 3.2: Descriptive statistics — explanatory variables, by variety

Variable	Unit	N	Mean	SD	Min	Max
Arabica						
Sugar	USD/lb	211	0.1692	0.0517	0.0886	0.3209
Cocoa	USD/tonne	211	2,572	469	1,482	4,215
CPI	index	211	241.9	27.0	201.5	307.8
WTI	USD/barrel	211	72.56	22.40	16.55	133.88
Fed Funds Rate	%	211	1.33	1.77	0.05	5.33
DXY	index	211	89.08	9.58	71.80	112.12
Production	thousand 60-kg bags	211	154,735	15,200	124,388	176,549
Consumption	thousand 60-kg bags	211	150,100	14,038	124,352	168,754
Exports	thousand 60-kg bags	211	127,226	14,126	101,361	144,896
Ending stocks	thousand 60-kg bags	211	33,574	5,370	23,121	43,132
Precipitation	mm	211	155.1	21.2	106.4	203.0
Temperature	°C	211	23.92	0.59	22.36	25.43
ONI	°C anomaly	211	−0.03	0.89	−1.64	2.75
COT net MM	contracts	211	2,168	31,708	−107,864	55,406
Robusta						
Sugar	USD/lb	192	0.1751	0.0502	0.1005	0.3209
Cocoa	USD/tonne	192	2,653	409	1,903	4,215
CPI	index	192	245.5	25.6	210.2	307.8
WTI	USD/barrel	192	72.82	23.20	16.55	133.88
Fed Funds Rate	%	192	0.96	1.37	0.05	5.33
DXY	index	192	89.78	9.72	71.80	112.12
Production	thousand 60-kg bags	192	157,386	13,169	129,774	176,549
Consumption	thousand 60-kg bags	192	152,363	12,610	125,867	168,754
Exports	thousand 60-kg bags	192	129,577	12,527	104,434	144,896
Ending stocks	thousand 60-kg bags	192	33,632	5,589	23,121	43,132
Precipitation	mm	192	155.1	21.5	106.4	203.0
Temperature	°C	192	23.94	0.61	22.36	25.43
ONI	°C anomaly	192	−0.01	0.90	−1.64	2.75
COT net MM	contracts	192	2,028	32,746	−107,864	55,406

3.4 Correlation Structure of the Predictor Set

The correlation matrix below covers the 2008–2023 sub-period, the longest interval over which every variable, including Robusta prices and the COT net position, is simultaneously available. Pearson coefficients are reported, consistent with the linear benchmark models estimated in Section 4.

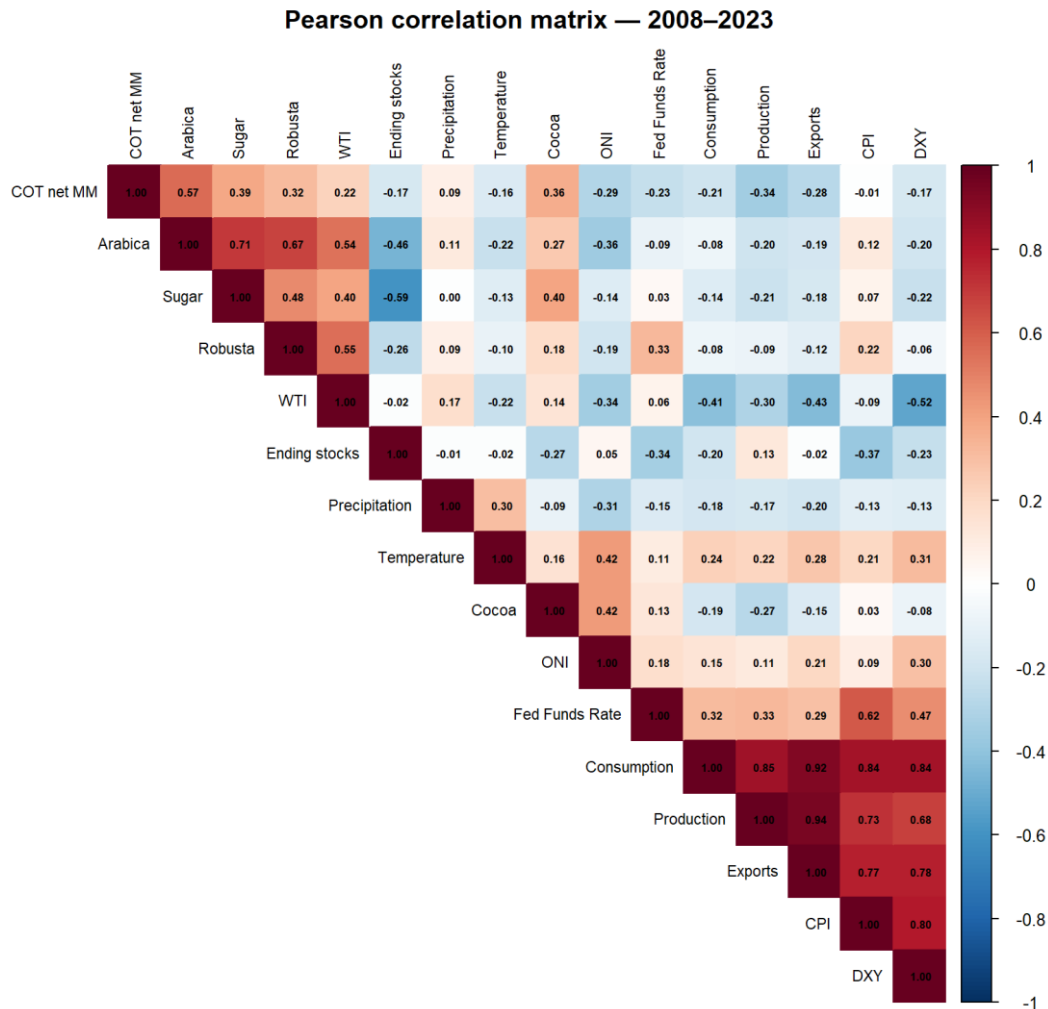


Figure 3.2: Pearson correlation matrix, 2008–2023

Among the price variables, Arabica and Sugar show the strongest pairwise correlation (0.71), while Arabica and Robusta sit more moderately at 0.67. The matrix’s most salient feature is the tight cluster formed by Production, Consumption, Exports and CPI, with pairwise coefficients ranging from 0.73 to 0.94, a direct reflection of their shared upward trends over the period and a first indication of the multicollinearity diagnosed via VIF in the Appendix.

Arabica prices carry a moderate negative correlation with Ending stocks (-0.46) and with ONI (-0.36), while its correlation with the COT net position (0.57) stands out as one of the strongest observed between any non-price predictor and a price variable. WTI and DXY are notably negatively correlated (-0.52). Climate variables, by contrast, show relatively weak correlations with prices in levels, generally staying below 0.20 in absolute value.

3.5 Estimation Strategy

3.5.1 Model Overview

Six forecasting models are estimated for each of the two dependent variables. Two econometric models serve as interpretable benchmarks: a linear regression (MLR) and an ARIMAX specification. Four machine learning models are estimated alongside: Random Forest, XGBoost, Support Vector Machine (SVM) and Artificial Neural Network (ANN). All models are estimated separately for Arabica and Robusta, reflecting the distinct market structures and price determinants of the two varieties. Given the structural data gaps documented in Section 3.1, the Arabica specifications are estimated over the 2006–2023 sub-period, from the point at which all fourteen exogenous regressors are simultaneously available, while all Robusta specifications are restricted to the 2008–2023 sub-period for which price data are available. Model performance is evaluated on three out-of-sample metrics: the coefficient of determination (R^2), Mean Squared Error (MSE) and Mean Absolute Error (MAE).

3.5.2 Walk-Forward Validation Protocol

All models are evaluated using a walk-forward expanding window validation protocol, which avoids the data leakage that would arise from conventional random cross-validation by ensuring that no future observations enter the training window at any estimation step. The dataset is split into a training set comprising the first 80% of observations and a test set

comprising the remaining 20%, yielding approximately 168 training and 43 test observations for the Arabica sample, and 153 training and 39 test observations for the Robusta sample.

The walk-forward procedure expands the training window by one observation at each step, re-estimates the model, and generates a one-step-ahead forecast. This procedure replicates the information set available to a forecaster at each point in time, using all observations up to time t to forecast the price at time $t + 1$, thereby ensuring that performance metrics reflect genuine out-of-sample predictive accuracy. Performance metrics are computed on the out-of-sample forecasts accumulated over the full test period, defined as follows:

$$R^2 = 1 - \frac{\sum_{t=1}^T (\hat{y}_t - y_t)^2}{\sum_{t=1}^T (\bar{y} - y_t)^2}$$

$$MSE = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|$$

where y_t is the observed price at time t , $\hat{y}_t$ is the model forecast, $\bar{y}$ is the mean of observed values over the test period rather than the training sample, so that R^2 is benchmarked against the variance actually realised during the forecast period, and T is the number of test observations.

3.5.3 Data Preprocessing and Stationarity

Prior to model estimation, all series are tested for stationarity using the Augmented Dickey-Fuller and KPSS tests, following the procedure detailed in the Appendix.

For the ARIMAX model, non-stationarity is handled internally by the automatic ARIMA selection procedure, which chooses the appropriate differencing order at each

estimation step. For the machine learning models, all variables are included in levels, as this allows the ML models to exploit the trending co-movement among variables that carries predictive information about the price trajectory, at the acknowledged cost of a potential spurious regression concern discussed in the limitations section.

Prior to ML model estimation, all variables are normalised to the $[0, 1]$ range using Min-Max scaling, ensuring that variables measured in different units and at different scales contribute equally to the model estimation and that numerical stability is maintained across all four ML algorithms:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

It should be noted that scaling parameters are derived from the full available sample for each variable rather than from the training set alone. This constitutes a minor form of look-ahead bias, as the minimum and maximum values of future observations are implicitly used when scaling early training windows. This simplification is acknowledged as a limitation of the present study. Predictions from the four machine learning models are converted back to their original price scale before evaluation, using the inverse of this Min-Max transformation.

3.5.4 ARIMAX

The ARIMAX model extends the standard ARIMA framework by incorporating a vector of exogenous regressors alongside the autoregressive and moving average components. Rather than re-selecting the (p, d, q) order at every step of the walk-forward loop, the order is selected once on the initial training window and held fixed for the remainder of the procedure. This order is chosen by minimising the corrected Akaike Information Criterion (AICc) over the initial training window, using the automatic selection procedure implemented in the forecast package in R, with a widened and exhaustive search: a maximum autoregressive order

of 5, a maximum moving average order of 5, and a maximum differencing order of 2, with both the stepwise and approximation shortcuts disabled to ensure the full candidate space is evaluated. Since this search is performed only once per series rather than at every one of the approximately 40 walk-forward steps, the wider and more exhaustive search remains computationally feasible. Seasonal terms are excluded from this search space: exploratory inspection of monthly average prices across the sample period showed no material seasonal pattern for either variety, consistent with Deina et al.'s (2022) independent finding of trend- and-cyclic but non-seasonal behaviour in separate Arabica and Robusta price series.

Once selected, the order is held fixed for the remainder of the walk-forward loop: at each step, only the coefficients of the fixed-order specification are re-estimated on the expanding training window via maximum likelihood, rather than re-running the automatic order-selection procedure. This constitutes a test of generalisation, asking whether an order chosen from the initial sample continues to fit well as new, unseen data are incorporated, rather than allowing the model structure itself to adapt at every forecast origin, which would blur the distinction between genuine forecasting and in-sample re-fitting of the model. The 14 predictors defined in Section 3.3 enter as contemporaneous exogenous regressors at each estimation step.

3.5.5 Linear Regression

The linear regression benchmark is estimated by ordinary least squares at each step of the walk-forward loop, re-estimating the model on the expanded training window before generating each one-step-ahead forecast. Unlike ARIMAX, the MLR specification includes no autoregressive component and makes no correction for serial correlation in the residuals, a limitation that is discussed further in the limitations section. The model is estimated on the levels dataset with the 14 exogenous predictors as regressors.

Ridge and LASSO regularised regression are also estimated as a robustness extension, with full methodology and results reported in the Appendix.

3.5.6 Random Forest

The Random Forest model grows an ensemble of 500 regression trees at each walk-forward step, with each tree trained on a bootstrap sample of the training observations and a random subset of predictors considered at each split and is re-estimated from scratch on the expanded training window at each forecast origin. The number of trees is set to 500 to ensure prediction stability while remaining computationally feasible within the walk-forward framework. The ensemble prediction is the average across all 500 trees, which reduces variance relative to any single tree without increasing bias. Variable importance is measured by the percentage increase in mean squared error when each predictor is randomly permuted while all others are held fixed, providing a model-agnostic indicator of predictor relevance that does not depend on the linearity of the relationships.

The sensitivity of Random Forest to tree count, complexity, and cost-complexity pruning is tested as a robustness extension, with full methodology and results reported in the Appendix.

3.5.7 XGBoost

The XGBoost model implements gradient boosting on regression trees, sequentially fitting each new tree to the residuals of the ensemble at the previous iteration. Hyperparameters, the number of boosting rounds, the maximum tree depth, the learning rate, and the row subsampling fraction, are selected via a time series cross-validation procedure conducted once on the initial training window for each series, rather than fixed manually. For each candidate combination, the model is trained and evaluated on three chronological splits of the training

window, using 60%, 70% and 80% of the window as training data respectively, with the following 20% held out as validation in each case; the combination minimising the average validation mean squared error across the three splits is retained. Arabica is searched over a finer grid centred on values from preliminary experiments (nrounds between 60 and 125, maximum depth between 3 and 5, learning rate between 0.06 and 0.12, and subsampling fraction between 0.75 and 0.9), while Robusta is searched over a broader grid (nrounds between 50 and 200, maximum depth between 3 and 6, learning rate between 0.05 and 0.3, and subsampling fraction between 0.7 and 1.0), reflecting the absence of a strong prior on the optimal range for the shorter Robusta series. The procedure selects 90 rounds, maximum depth 4, learning rate 0.08, and subsampling fraction 0.8 for Arabica, and 100 rounds, maximum depth 3, learning rate 0.3, and subsampling fraction 0.9 for Robusta, reflecting a slower and more conservative learning process for Arabica against a faster, shallower specification for Robusta.

3.5.8 Support Vector Machine

The Support Vector Machine is estimated with a radial basis function kernel, which maps inputs into an implicit high-dimensional feature space and allows the model to capture non-linear relationships between predictors and price levels. The model is re-estimated from scratch on the expanded training window at each forecast origin. The regularisation parameter (cost) and the kernel bandwidth (gamma) are selected via the same three-split time series cross-validation procedure used for XGBoost, evaluating 30 combinations (cost between 0.1 and 10, gamma between 0.01 and 0.5) against the average validation mean squared error across the three chronological splits. For Robusta, this procedure selects cost = 10 and gamma = 0.05, improving on the values used in the original specification. For Arabica, every tuned combination underperforms the original specification (cost = 5, gamma = 0.05) on the test set, so these values are retained as the best configuration identified across the tuning exercise.

3.5.9 Artificial Neural Network

The ANN is implemented as a single hidden layer feedforward network with a sigmoid activation function in the hidden layer and a linear activation function on the output neuron, the latter required for continuous regression targets. The number of hidden neurons (size) and the L2 regularisation strength (decay) are selected jointly via the same three-split time series cross-validation procedure used for XGBoost and SVM, evaluating 24 combinations, size among 2, 3, 5, 8, 10 and 15 neurons, decay among 0, 0.001, 0.01 and 0.1, against the average validation mean squared error across the three chronological splits. This candidate range brackets the five-neuron architecture Kohzadi et al. (1996) found optimal for a comparable monthly commodity-price forecasting problem. The procedure selects a two-neuron architecture for both series, with a decay of 0.1 for Arabica and 0.001 for Robusta, indicating that Arabica benefits from substantially stronger regularisation than Robusta. Both series therefore converge to the smallest architecture tested, consistent with the limited size of the available training sample. The model is re-estimated from scratch on the expanded training window at each forecast origin, trained by backpropagation for a maximum of 500 iterations at each step. A fixed random seed is applied before each candidate configuration is fitted during tuning, and again before the walk-forward loop itself, to ensure reproducibility across runs.

Network depth and activation function are also tested as robustness extensions, with full methodology and results reported in the Appendix.

3.5.10 Model Interpretation: Permutation Importance and Partial Dependence

The six models compared in this study differ substantially in how interpretable their parameters are. The MLR and ARIMAX coefficients can be read directly, but the four ML models, Random Forest, XGBoost, SVM and ANN, do not expose comparable parameters: a tree ensemble or a support vector machine does not have a single coefficient per predictor that

can be examined the way an OLS coefficient can. Two model-agnostic techniques are used to address this gap, applied identically across all four ML models so that their outputs remain directly comparable.

Permutation importance quantifies how much each model relies on a given predictor for its out-of-sample accuracy. For each of the 14 predictors, the corresponding column in the test set is randomly shuffled while every other predictor is left unchanged, the model generates new predictions on this perturbed test set, and the resulting increase in MSE relative to the model's actual walk-forward forecasts is recorded as that predictor's importance score. A larger increase indicates that the model depends more heavily on the correct, unshuffled values of that variable; a score close to zero or negative indicates that the variable contributes little, or that shuffling it does not meaningfully degrade the forecasts. This procedure is applied to a single instance of each model fitted on the full initial training window, separately for Arabica and Robusta. For ARIMAX, only the exogenous regressors are permuted, with the autoregressive component left untouched, since the autoregressive structure has no direct counterpart among the 14 predictors. MLR is included for completeness, though its importance scores are affected by multicollinearity among the predictors, a problem discussed in detail in the results.

Partial dependence complements this by showing the direction and shape of each predictor's relationship with the forecasted price, rather than only its overall contribution to accuracy. For a given predictor, a grid of values spanning its range as observed in the training set is constructed, and at each grid point that predictor is set to the same fixed value across every training observation while all other predictors are left at their actual, observation-specific values. The model then generates a prediction for every modified observation, and these predictions are averaged to produce a single point on the partial dependence curve. Repeating this across the full grid traces out how the model's average prediction moves as that one

Predicting Arabica and Robusta Coffee Futures Prices

predictor changes, with the influence of the remaining predictors averaged out. The grid is restricted to the range of values actually observed in the training set, so the model is never evaluated outside the range of data it was trained on.

4. Results

This chapter presents the out-of-sample results for the six models introduced in Chapter 3. Section 4.1 reports aggregate performance across all six models and both varieties, using R^2 as the primary comparison metric, together with the forecast trajectories over the test period. Section 4.2 turns to interpretation, first through the coefficient-based reading available for MLR and ARIMAX, then through the variable importance and partial dependence results computed for the four machine learning models.

4.1 Model Performance

R^2 is used as the primary basis for comparison across models and varieties throughout this chapter, since it is scale-invariant. MSE and MAE, by contrast, are not directly comparable across model families: ARIMAX and MLR are estimated on the original price scale, while the four machine learning models are estimated on Min-Max scaled $[0, 1]$ data, so their MSE and MAE values are only meaningful within a single model and variety, not across them. Table 4.1 reports all three metrics for every model and variety combination for completeness.

Table 4.1: Comparative model performance — out-of-sample results

Model	R^2	MSE	MAE
Arabica			
MLR	0.7836	0.0352	0.1535
ARIMAX	0.9547	0.0074	0.0655
Random Forest	0.8152	0.0082	0.0664
XGBoost	0.8117	0.0084	0.0676
SVM	0.8130	0.0083	0.0564
ANN	0.7282	0.0121	0.0924
Robusta			
MLR	0.2759	114285	274.6
ARIMAX	0.9257	11724	89.8
Random Forest	0.6759	0.0214	0.1111
XGBoost	0.7970	0.0134	0.0874
SVM	0.7607	0.0158	0.0995
ANN	0.4560	0.0360	0.1491

Predicting Arabica and Robusta Coffee Futures Prices

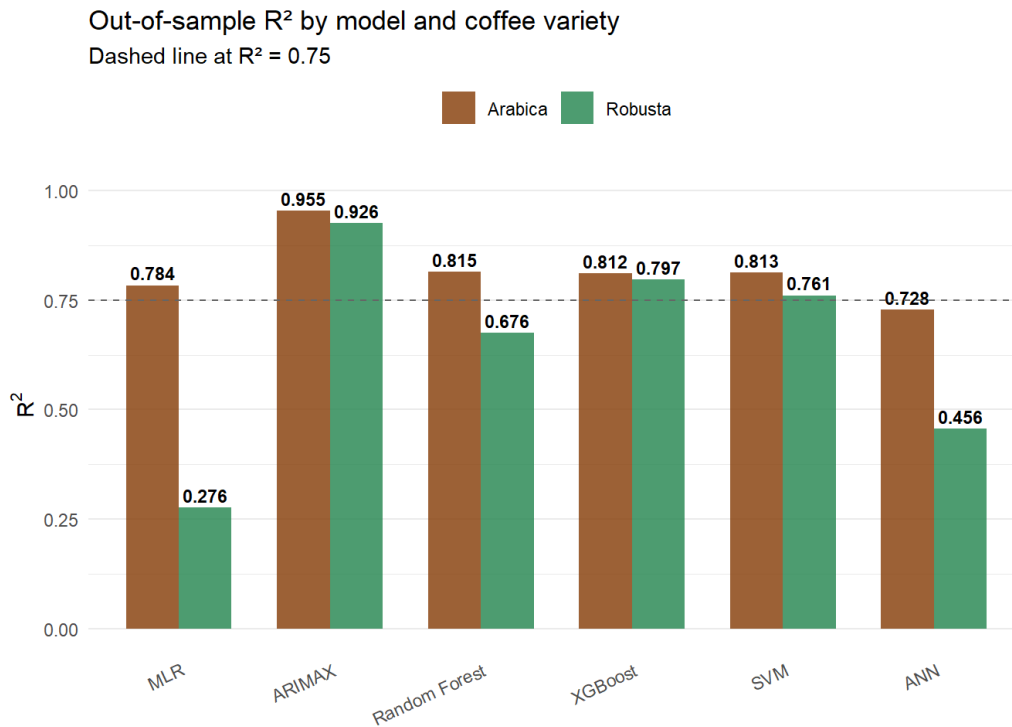


Figure 4.1: Out-of-sample R^2 by model and coffee variety

ARIMAX is clearly the strongest model for both varieties, reaching 0.955 for Arabica and 0.926 for Robusta, well ahead of every other model. Among the machine learning models, XGBoost, SVM and Random Forest sit close together for Arabica, all around 0.81. Robusta pulls them apart: XGBoost holds up best at 0.797, SVM follows at 0.761, and Random Forest falls further behind at 0.676. The ANN is the weakest of the four machine learning models for both varieties, at 0.728 and 0.456. MLR performs reasonably for Arabica (0.784) but produces the weakest result of all six models for Robusta, at 0.276. Because every figure comes from the same walk-forward protocol, with no model ever trained on data it is later asked to predict, the comparison is a fair one: most models clear an R^2 of 0.75, with four exceptions, MLR and Random Forest for Robusta, and the ANN for both varieties.

4.1.1 Adjustment Quality

Beyond the aggregate R^2 reported in Table 4.1, adjustment quality can be assessed at the level of each model's own fitting choices. Random Forest, evaluated at its default configuration, reaches an R^2 of 0.815 for Arabica and 0.676 for Robusta. XGBoost, SVM and

Predicting Arabica and Robusta Coffee Futures Prices

the ANN are each tuned via three-split time series cross-validation; Table 4.2 reports the selected configuration for each. Hyperparameter sensitivity analyses for all four models are reported in the Appendix.

For the two econometric models, adjustment quality also extends to the calibration of their 95% prediction intervals: empirical coverage is close to nominal for MLR-Arabica (97.7%) but falls short elsewhere (81.4%-84.6%), a pattern linked to the non-stationarity limitation discussed in Chapter 6.

Table 4.2: Selected hyperparameters by model and variety

Model	Arabica	Robusta
Random Forest	500 trees (default)	500 trees (default)
XGBoost	90 rounds, depth 4, $\eta = 0.08$, subsample 0.8	100 rounds, depth 3, $\eta = 0.30$, subsample 0.9
SVM	cost = 5, $\gamma = 0.05$	cost = 10, $\gamma = 0.05$
ANN	2 neurons, decay = 0.1	2 neurons, decay = 0.001

4.1.2 Forecast Trajectories

Figure 4.2 and Figure 4.3 plot the walk-forward forecasts against observed prices for Arabica and Robusta respectively, across all six models, with a 95% prediction interval shown for the two econometric models, ARIMAX and MLR.

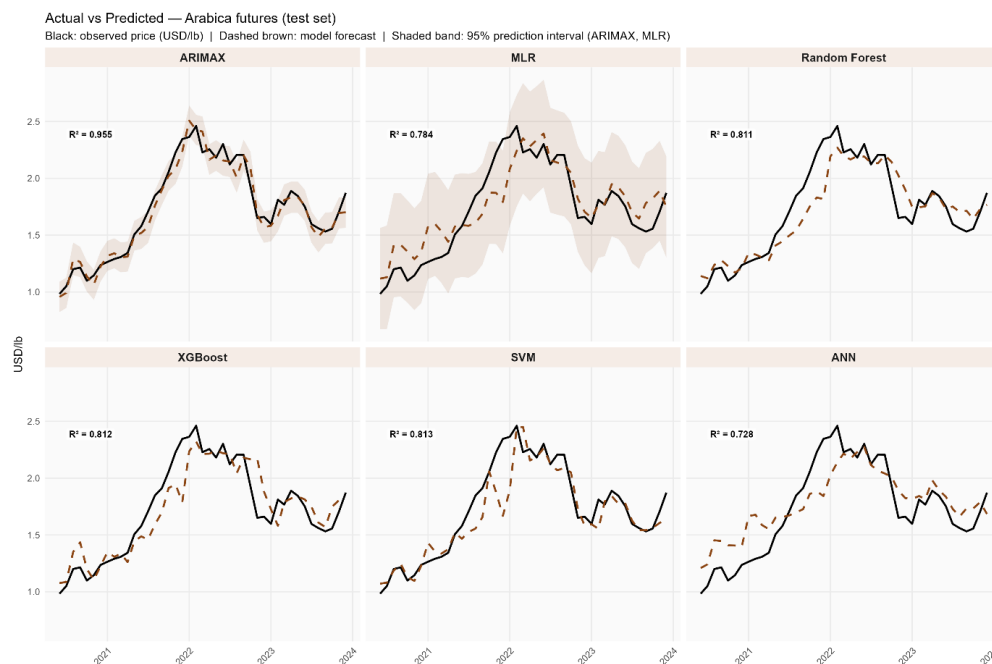


Figure 4.2: Actual vs predicted — Arabica futures (test set)

Predicting Arabica and Robusta Coffee Futures Prices

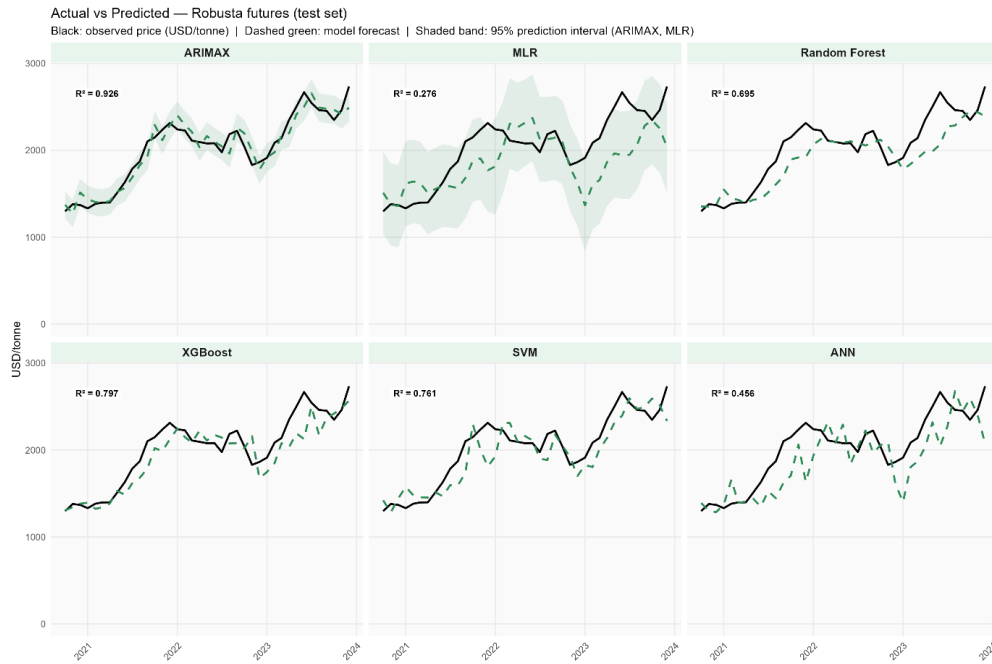


Figure 4.3: Actual vs predicted — Robusta futures (test set)

ARIMAX tracks the observed series closely for both varieties, including through the sharp Arabica peak of 2021–2022 and the Robusta acceleration from 2023 onward, with a comparatively narrow prediction interval reflecting its strong fit. MLR tracks Arabica reasonably well but fails visibly for Robusta, its forecasts oscillating erratically within a much wider prediction interval that reflects the instability discussed in Section 4.2.1. XGBoost, Random Forest and SVM capture the broad direction but lag at turning points and compress the amplitude of short-term movements, a typical feature of one-step-ahead forecasts. The ANN follows the general trajectory but remains the most unstable of the six models, particularly for Robusta.

Predicting Arabica and Robusta Coffee Futures Prices

Figure 4.4 and Figure 4.5 show the corresponding forecast residuals.

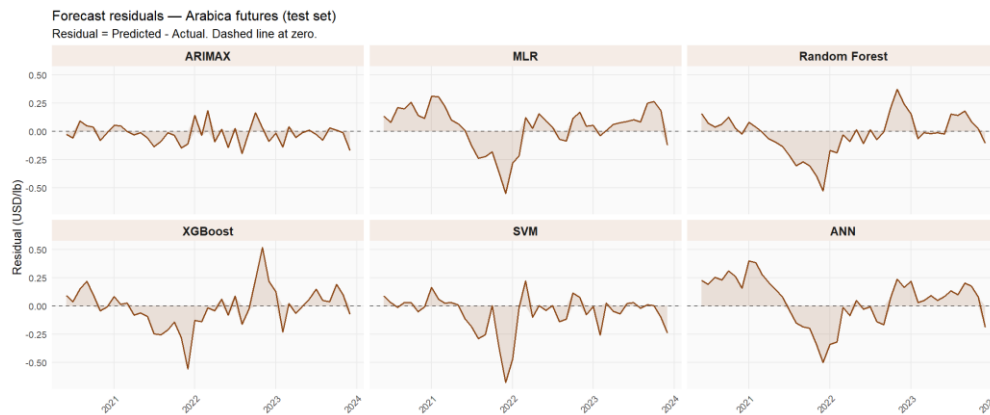


Figure 4.4: Forecast residuals — Arabica futures (test set)

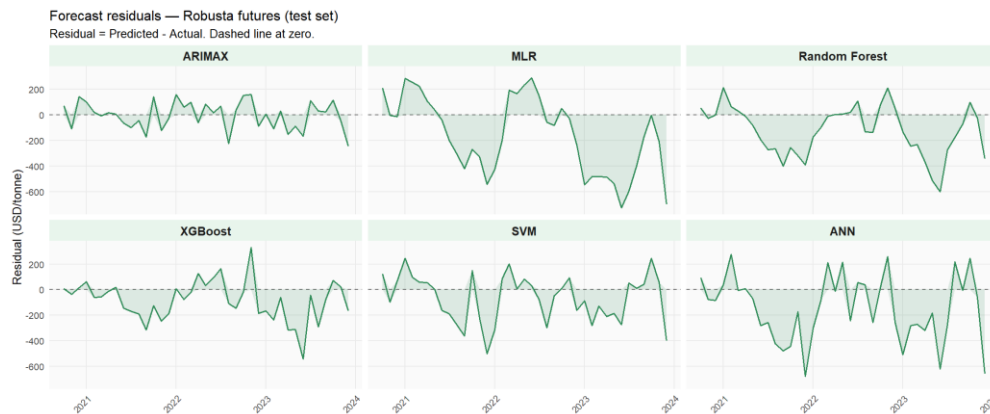


Figure 4.5: Forecast residuals — Robusta futures (test set)

For Arabica, most models show a pronounced negative spike just before the 2022 peak, reflecting a common failure to anticipate the price surge; XGBoost and Random Forest carry larger positive residuals from late 2022 onward, suggesting they were slow to incorporate the subsequent decline. For Robusta, ARIMAX residuals oscillate around zero with no persistent bias, while Random Forest, MLR and XGBoost each show extended negative episodes tied to the 2021–2023 period, and the ANN residuals remain the most erratic throughout.

4.2 Model Interpretation

This section turns from aggregate accuracy to what each model says about the predictors themselves. Section 4.2.1 reads the coefficients of the two econometric models directly, since both admit a direct economic interpretation, before Section 4.2.2 turns to variable importance

and partial dependence, the two techniques used to interpret the four machine learning models instead.

4.2.1 Econometric Models

Linear Regression and ARIMAX are treated together in this section because their coefficients admit a direct economic reading, unlike the four machine learning models discussed in the following section. Table 4.3 and Table 4.4 report the ordinary least squares and ARIMAX coefficients for the 14 predictors, estimated on the full initial training window for each variety, with statistical significance indicated by conventional significance stars.

Table 4.3: MLR and ARIMAX regression coefficients — Arabica

Variable	MLR		ARIMAX	
	Estimate	p-value	Estimate	p-value
Sugar	+2.5872	0.0001 ***	+1.0204	0.0021 **
Cocoa	+0.0000	0.8550	+0.0000	0.8565
CPI	−0.0050	0.4320	+0.0024	0.5435
WTI	+0.0068	0.0005 ***	+0.0018	0.0009 ***
Fed Funds Rate	−0.0032	0.8648	−0.0579	0.0000 ***
DXY	−0.0045	0.5111	−0.0023	0.5098
Production	+0.0000	0.6259	−0.0000	0.9636
Consumption	+0.0000	0.1037	+0.0000	0.6206
Exports	+0.0000	0.7168	−0.0000	0.9256
Ending stocks	−0.0000	0.0024 **	+0.0000	0.8800
Precipitation	+0.0010	0.3205	+0.0001	0.6152
Temperature	−0.0375	0.3204	+0.0114	0.5360
ONI	−0.0070	0.8121	+0.0330	0.3381
COT net MM	+0.0000	0.0000 ***	+0.0000	0.0250 *

Table 4.4: MLR and ARIMAX regression coefficients — Robusta

Variable	MLR		ARIMAX	
	Estimate	p-value	Estimate	p-value
Sugar	+1595.9308	0.0259 *	+317.4372	0.4832
Cocoa	−0.2270	0.0025 **	−0.0128	0.8011
CPI	−20.7537	0.0022 **	+14.1356	0.0014 **
WTI	+11.2848	0.0000 ***	+2.5366	0.0235 *
Fed Funds Rate	+46.0073	0.1754	+35.6851	0.4112
DXY	+15.3865	0.0304 *	−3.9373	0.0905 .
Production	−0.0116	0.1474	−0.0023	0.6194
Consumption	+0.0282	0.0001 ***	−0.0008	0.8581
Exports	+0.0083	0.4090	−0.0023	0.6749
Ending stocks	+0.0184	0.0160 *	−0.0085	0.1371
Precipitation	+1.4826	0.1974	+0.2623	0.3572
Temperature	+0.5158	0.9898	−18.0120	0.2179
ONI	−49.5998	0.1464	−6.6544	0.8677
COT net MM	+0.0035	0.0004 ***	+0.0029	0.0000 ***

For Arabica, the two models agree on COT net Managed Money and WTI as significant, positively signed predictors, consistent with the expectation that bullish speculative positioning

and rising energy costs align with rising coffee prices, and on Sugar, also significant and positively signed under both specifications. Beyond this common ground, the two models diverge: MLR additionally finds Ending stocks significant and negative, consistent with inventory theory, while ARIMAX instead finds the Federal Funds Rate significant and negative. Four of the 14 predictors reach conventional significance under each model, though not entirely the same four. Under MLR, two coefficients carry the wrong sign relative to economic theory, CPI and Production, and both are statistically insignificant.

For Robusta, both models again agree on WTI and COT net Managed Money as significant, positively signed predictors. MLR additionally finds Sugar and Consumption significant with the expected sign, but four further coefficients, CPI, DXY, Cocoa and Ending stocks, are significant with the wrong sign. Under ARIMAX, CPI is significant as well but with the theoretically consistent positive sign, DXY approaches significance at the 10% level with the expected negative sign, and Cocoa and Ending stocks are not significant. This pattern, where several of MLR's wrong-signed Robusta coefficients turn correctly signed or insignificant under ARIMAX, is consistent with ARIMAX's autoregressive component absorbing part of the shared trend among the predictors, reducing the burden placed on the exogenous coefficients relative to MLR's purely cross-sectional specification.

4.2.2 Variable Importance and Partial Dependence

Permutation importance and partial dependence plots are used to interpret the four machine learning models, following the procedure described in Section 3.5.10. MLR is excluded from this comparison because severe multicollinearity produces negative importance scores throughout; ARIMAX is excluded because it predicts on the original price scale rather than the Min-Max scaled range used by the four ML models, making its scores not directly comparable. Full rankings for all 14 predictors and per-model partial dependence detail are reported in the Appendix.

Predicting Arabica and Robusta Coffee Futures Prices

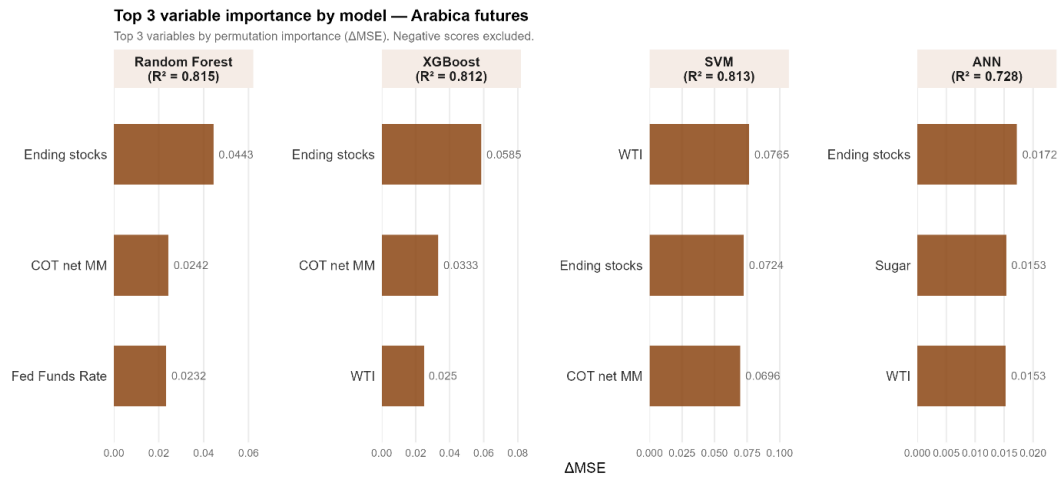


Figure 4.6: Top 3 variable importance by model — Arabica

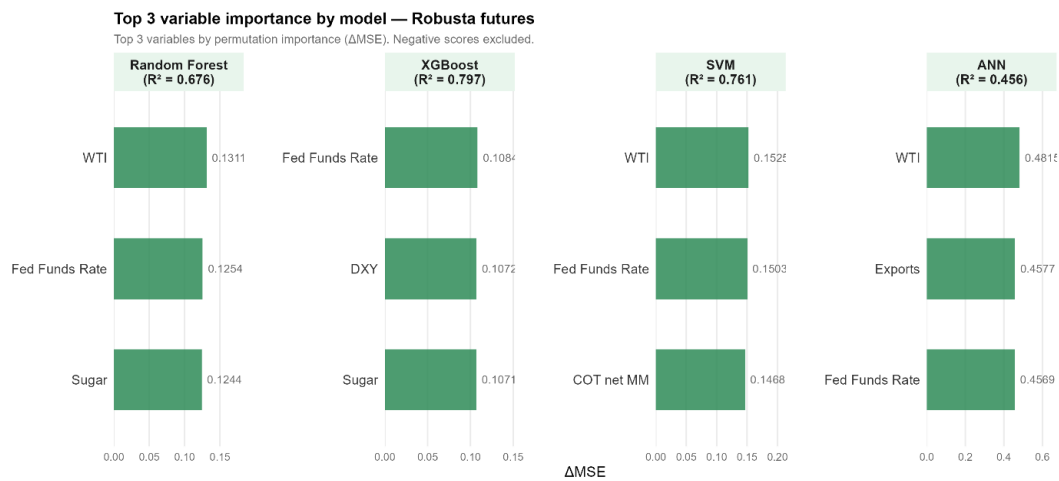


Figure 4.7: Top 3 variable importance by model — Robusta

For Arabica, Ending stocks ranks among the top three predictors across all four models, the most consistent finding in the comparison; COT net Managed Money and WTI also feature prominently. For Robusta, macroeconomic variables dominate instead: WTI ranks first for three of the four models, and the Federal Funds Rate appears in the top three of all four, making it the most consistently important predictor for this variety.

Partial dependence plots complement permutation importance by showing the direction and shape of each predictor's relationship with the forecast, restricted to the consensus drivers identified above: Ending stocks, COT net Managed Money and WTI for Arabica, and the Federal Funds Rate and WTI for Robusta.

Predicting Arabica and Robusta Coffee Futures Prices

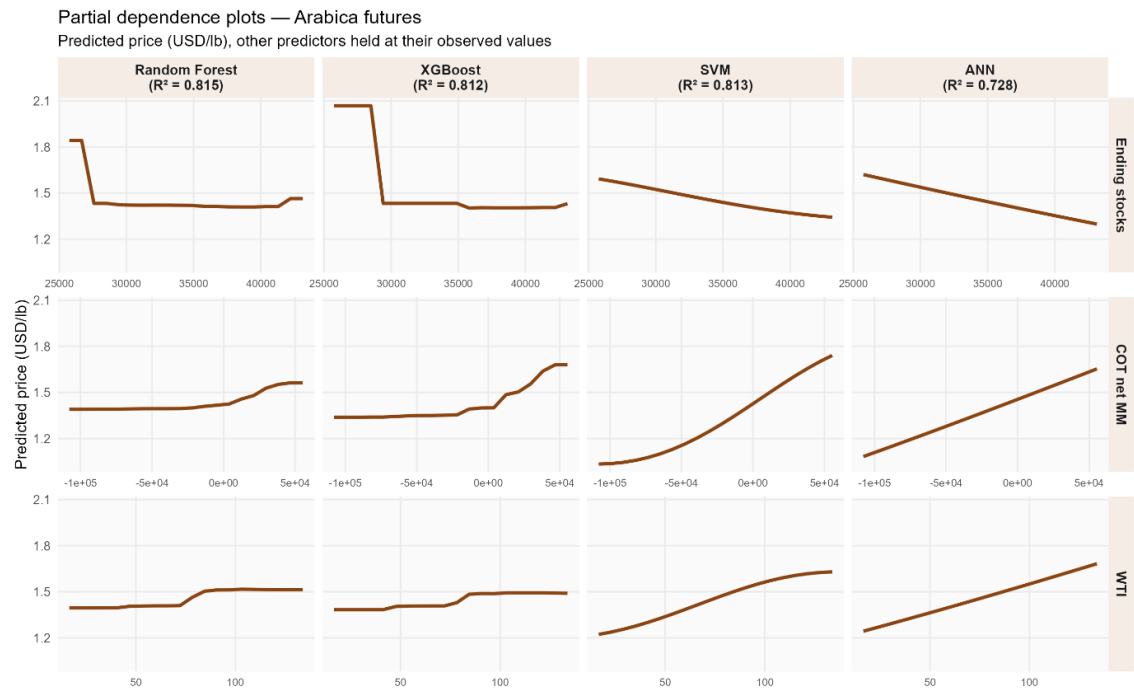


Figure 4.8: Partial dependence plots — Arabica futures

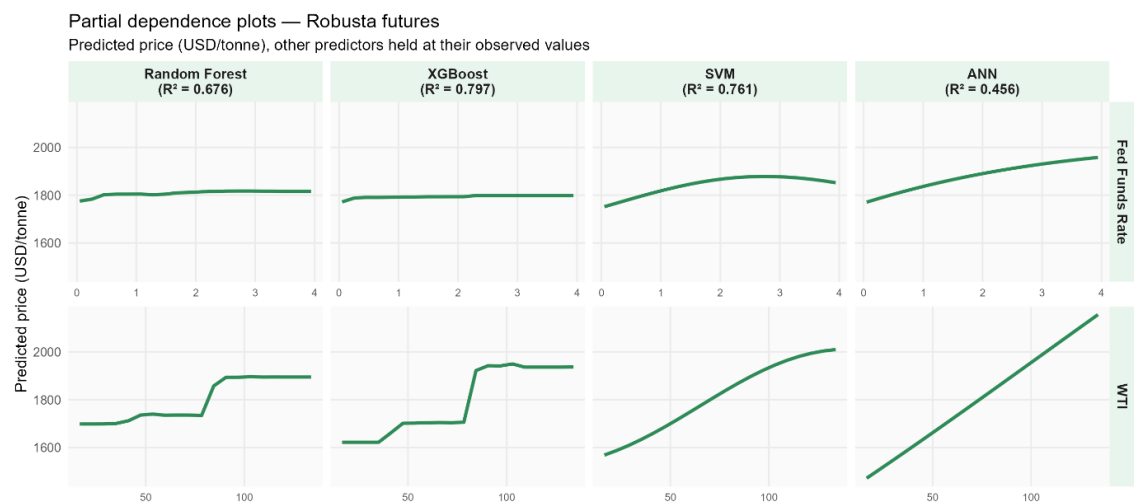


Figure 4.9: Partial dependence plots — Robusta futures

For Arabica, all four models agree on direction: Ending stocks pulls the forecast down consistently, while COT net Managed Money and WTI push it up, consistent with inventory theory, the financialisation argument, and the energy cost channel respectively. For Robusta, WTI is unambiguous again, rising across all four models, but the Federal Funds Rate is messier and none of the four models reproduces the negative relationship that monetary policy theory would suggest. This is consistent with the non-stationarity limitation discussed in Chapter 6: the Federal Funds Rate rose sharply from 2022 onward, followed by Robusta's own price

acceleration from 2023, so a positive rather than a negative relationship may simply reflect two upward-trending series over overlapping years rather than a causal monetary policy channel.

5. Discussion

5.1 Overall Model Performance

Returning to the research questions posed in Chapter 1, this section addresses model accuracy and the comparison between econometric and machine learning approaches (RQ1, RQ2); Sections 5.2 and 5.3 turn to the determinants driving that accuracy and how they differ between Arabica and Robusta (RQ3, RQ4). The results in Chapter 4 point to two broad conclusions. ARIMAX's explicit autoregressive structure gives it a clear edge over every other model, for both varieties, confirming that price persistence is a real source of short-run predictability that the five remaining models, relying solely on contemporaneous exogenous predictors, cannot fully replicate. The second conclusion is about spread rather than level: the gap between the strongest and weakest models is much wider for Robusta than for Arabica, for reasons Section 5.3 returns to.

For Arabica, ARIMAX stands clearly apart at $R^2 = 0.955$ (Table 4.1). XGBoost, Random Forest and SVM form a tight second tier around 0.81, close enough to suggest the three are picking up much the same signal rather than exploiting genuinely different information. MLR reaches 0.784, and the ANN brings up the rear at 0.728 (Chapter 6 returns to why).

For Robusta, ARIMAX again leads, at $R^2 = 0.926$, though the margin over XGBoost (0.797), SVM (0.761) and Random Forest (0.676) is narrower than for Arabica. The standout result, for the wrong reasons, is MLR's collapse to $R^2 = 0.276$, with forecasts that oscillate erratically (Figure 4.3). OLS is known to struggle with multicollinearity in short samples, and that is exactly what is happening here: with only 153 training observations and 14 collinear predictors, individual coefficients carry high variance, so a small change in the expanding training window can shift them substantially from one walk-forward step to the next. The regularisation exercise reported in the Appendix backs this up directly: Ridge nearly doubles

Robusta's R^2 to 0.464 while barely touching Arabica's, precisely the asymmetry a multicollinearity story would predict.

The ARIMAX coefficient tables (Table 4.3, Table 4.4) show the same problem from another angle: only a handful of regressors reach conventional significance for either variety, not because the others carry no information, but because Production, Consumption, Exports and CPI all trend together over the sample (VIF above 20, reported in the Appendix), so OLS cannot pull apart their individual contributions. If anything, this makes a retrospective case for the machine learning models used throughout this study, which are built to exploit exactly this kind of signal, spread across a correlated group of variables, that linear regression cannot isolate. One caveat carries through the rest of this chapter, however: these R^2 figures come from series in levels, which can inflate R^2 beyond what a stationary series would deliver, a point taken up fully in Chapter 6.

This overall pattern, where ARIMAX outperforms every machine learning model for both varieties, sits in some tension with the wider literature, though not with all of it. Kohzadi et al. (1996), discussed in Section 2.4, found a neural network outperforming an ARIMA benchmark for wheat and live cattle prices, and Deina et al. (2022) found non-linear methods outperforming linear benchmarks consistently across both coffee varieties and all forecast horizons tested. Here, the reverse holds for every machine learning model, including the ANN, a pattern Masini et al. (2023), reviewed in Section 2.4, also observe in their own empirical illustration, where a penalised linear regression outperforms random forest. ARIMAX's advantage here is therefore consistent with part of the wider evidence base, not simply an anomaly specific to this thesis, even if it diverges from Kohzadi's and Deina's coffee- and commodity-specific results. The exogenous predictor set built into ARIMAX throughout this thesis, richer than a purely autoregressive benchmark, is one plausible difference from Kohzadi's and Deina's comparisons specifically, though this thesis does not test that distinction

directly. Nayak et al. (2024) offer indirect support for this reading. Comparing an almost identical model set, ARIMAX, MLR, ANN, SVR and Random Forest, for vegetable price forecasting in India, they find that an ARIMAX specification built with only two exogenous weather variables achieves a lower RMSE than all four alternatives for two of the three crops studied, tomato and potato, though the ordering reverses for the third, onion. This is not a direct test of a fourteen-variable specification, but it suggests that even a modest exogenous predictor set can be enough to close the gap reported in univariate comparisons such as Kohzadi's and Deina's.

5.2 The Role of Financial and Speculative Variables

The permutation importance results (Section 4.2.2), run consistently across all four machine learning models, give the clearest picture of what actually drives out-of-sample accuracy. For Arabica, Ending stocks lands in the top three for every model, the single most consistent finding anywhere in this comparison. COT net Managed Money and WTI follow close behind. That speculative positioning matters this much fits the broader story of financialisation in commodity futures markets discussed in Section 2.2: the net position held by Managed Money participants captures expectations about where prices are heading before those expectations show up in the fundamentals themselves.

A further, more speculative possibility is worth raising: if the positive COT coefficient found in both MLR and ARIMAX, for both varieties (Table 4.3, Table 4.4), reflects funds buying in anticipation of rising prices, and that buying pressure helps push prices up in turn, the resulting rise could feed back into the original positioning rather than just confirming it. Nothing in this analysis tests that feedback loop directly, and the lagged COT check in the Appendix, which finds that adding lagged values does not meaningfully improve most forecasts, only speaks to it indirectly: whatever signal past positioning carries appears already priced into the current figure.

Robusta tells a different story. The Federal Funds Rate lands in the top three for all four models, the strongest consensus result for this variety, and WTI comes first for three of the four. Energy costs mattering this much makes sense given where Robusta is actually grown: Vietnam alone accounts for over 40% of the weighted Robusta aggregate (see Appendix) and grows almost exclusively Robusta, so a cost shock there falls on this variety specifically rather than spreading across both. As for the Federal Funds Rate, its partial dependence shape (Figure 4.9) hints at why it matters so consistently: the rate sat near zero for most of 2008 to 2021, then climbed sharply from 2022, a shift that lines up almost exactly with the test period.

Two things stand out once both varieties are considered together. Neither Ending stocks nor the Federal Funds Rate, the two most consistently important predictors overall, is obviously ‘financial positioning’ in the narrow sense, which suggests the informative variables genuinely differ by variety rather than clustering into one dominant category. Climate variables, for both varieties, sit consistently in the lower half of the ranking, averaging 9.1 and 9.6 of 14 across the eight model-variety combinations. Rather than a disappointing result, this echoes Bastianin et al. (2018), discussed in Section 2.3, who find that ENSO shocks account for a modest share of the variance in Colombian coffee prices, with the much larger remainder attributed to broader world coffee market shocks, a category this thesis’s macroeconomic and financial predictors are well placed to capture. The full climate robustness analysis is reported in the Appendix alongside the other robustness checks.

5.3 Arabica vs Robusta: Structural Differences

Fixing the ARIMAX order flips the ranking between the two varieties. Arabica now comes out ahead ($R^2 = 0.955$, ARIMA(2,0,1)) rather than behind Robusta ($R^2 = 0.926$, ARIMA(1,0,3)), the opposite of what the re-adaptive specification produced (see Appendix). A single, thorough search on the initial window seems to have removed a source of noise that hit Arabica harder under the re-adaptive approach, likely because that approach sometimes

landed on an unstable order when the training window was still small, early in the walk-forward loop.

Why does MLR fail for Robusta but hold up for Arabica? Partly sample size, 153 training observations against 168, but Robusta's predictors are also more tangled with each other: Production, Consumption, Exports and CPI all trend together over the shorter 2008–2023 window. Ridge nearly doubling Robusta's R^2 while making Arabica's slightly worse points to a genuine difference in how entangled each variety's predictors are, not sample size alone.

Random Forest (0.676) and XGBoost (0.797) also do noticeably worse on Robusta than on Arabica (0.815 and 0.812), a structural limit of ensemble methods rather than anything specific to Robusta's predictors: both predict by averaging over training observations, so neither can extrapolate meaningfully beyond the range of prices seen in training. Robusta's sharp price acceleration from mid-2023 pushes past that range, and both models respond by flattening out while the actual series keeps climbing, the persistent negative bias visible in the residual plots (Figure 4.5) from late 2022 onward.

This divide between the two varieties is not one the existing literature anticipates. Section 2.3 explains why Arabica and Robusta diverge through climate, different agronomic zones and opposite ENSO exposure, but the results here point to a macro-financial divide instead: Robusta's sensitivity to WTI and the Federal Funds Rate has no established explanation in the literature reviewed for this thesis. Maurice and Davis (2012), cited in Section 2.2, find that oil price movements affect both varieties, but say nothing about why one might be more exposed than the other. The asymmetry is reported here as an empirical finding rather than a sourced mechanism.

6. Conclusion

This study set out to forecast monthly Arabica and Robusta coffee futures prices using six models, two econometric benchmarks (ARIMAX and Linear Regression) and four machine learning algorithms (Random Forest, XGBoost, Support Vector Machine and Artificial Neural Network), estimated under a walk-forward expanding window protocol with an 80/20 train/test split. All six were evaluated on the same three out-of-sample metrics, against the same predictor set spanning macroeconomic, fundamental, climatic and speculative variables.

ARIMAX comes out on top for both varieties, reaching $R^2 = 0.955$ for Arabica and 0.926 for Robusta, with XGBoost the strongest of the remaining models (0.812 and 0.797). MLR tells a cautionary tale: it performs reasonably for Arabica but collapses for Robusta, to $R^2 = 0.276$, a multicollinearity problem that Ridge regression nearly resolves, almost doubling Robusta's R^2 in the process.

The permutation importance results show that what drives forecasting accuracy is not the same for both varieties: Ending stocks, COT net Managed Money and WTI matter most for Arabica, while WTI and the Federal Funds Rate matter most for Robusta. Climate variables contribute comparatively little for either variety at a one-month horizon, a genuine but bounded role consistent with Bastianin et al. (2018).

The robustness checks and tuning extensions reported in the Appendix largely confirm these findings: disaggregating climate or adding GDP growth and lagged COT positioning does not improve results, while restricting predictors to each variety's own producing countries helps Robusta, pushing XGBoost to its strongest result anywhere in this study ($R^2 = 0.849$), but not Arabica.

Several limitations qualify these findings: non-stationarity in how the machine learning models are estimated, instability in the ANN across depth and activation function, a hyperparameter search that does not fully nest inside the walk-forward loop, climate indices

that take no account of within-country variation, data constraints that restrict each variety to a shorter estimation sample than the full 2000-2023 period, and the assumption that predictor relationships stay fixed across a period when the coffee market itself changed substantially.

These limitations suggest a few directions worth pursuing. Lagged climate variables at three, six and twelve months could capture the delayed weather-to-price channel that a contemporaneous specification cannot, though the lagged COT check in this study is a reminder that such extensions need to be tested, not assumed to help. A more systematic setup for the ANN, for example averaging results across several random initialisations, could address its instability more directly than any single fix tried here. And a rolling-window or regime-switching specification could let predictor relationships evolve over time, closer to how the coffee market actually behaved across the sixteen to eighteen years each model was actually estimated on.

From a practical standpoint, the results point toward inventory levels, speculative positioning, energy prices and, for Robusta specifically, the interest rate environment, as the more useful short-run signals for coffee, monitored together rather than any one in isolation; climate conditions at monthly frequency are the clearer exception, given the lag between a weather shock and its eventual effect on prices. For Robusta specifically, the variety-specific results suggest tracking supply conditions in its own producing countries, chiefly Vietnam and Brazil, the two largest Robusta producers worldwide, followed by Indonesia, India, Uganda and Côte d'Ivoire, rather than global aggregates that dilute the signal relevant to this variety. Taken together, these results point to a practical takeaway as much as a modelling one: which signals matter, and how they matter, differs enough between Arabica and Robusta that treating them as a single coffee market would miss real structure in the data.

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8. Appendices

8.1 Stationarity Testing: ADF and KPSS Procedures

Prior to model estimation, all series are tested for stationarity using two complementary procedures, referenced from Section 3.5.3. The Augmented Dickey-Fuller (ADF) test evaluates the null hypothesis of a unit root; rejection at the 5% level indicates stationarity. The KPSS test evaluates the null hypothesis of stationarity; rejection at the 5% level indicates non-stationarity. A series is classified as stationary when the ADF rejects the unit root hypothesis and the KPSS fails to reject the stationarity hypothesis. In cases of conflicting results, the series is tested in first differences; if both tests then agree on stationarity, the series is classified as integrated of order one.

8.2 MLR and ARIMAX: Coefficient Diagnostics and Robustness

This appendix reports the full coefficient diagnostics and regularisation checks for the two econometric models, MLR and ARIMAX, referenced from Section 4.2.1.

MLR

Coefficients with expected sign consistency. Table 8.1 and Table 8.2 report the OLS coefficients estimated on the full initial training window for each variety, alongside the sign that economic theory predicts, and flag any disagreement between the two.

Table 8.1: MLR regression coefficients with expected vs observed signs — Arabica

Variable	Estimate	t-stat	p-value	Expected sign	Consistent
Sugar	2.5872	4.04	0.0001 ***	+	Yes
Cocoa	0.0000	0.18	0.8550	+	Yes
CPI	-0.0050	-0.79	0.4320	+	No
WTI	0.0068	3.59	0.0005 ***	+	Yes
Fed Funds Rate	-0.0032	-0.17	0.8648	-	Yes
DXY	-0.0045	-0.66	0.5111	-	Yes
Production	0.0000	0.49	0.6259	-	No
Consumption	0.0000	1.64	0.1037	+	Yes
Exports	0.0000	0.36	0.7168	?	—
Ending stocks	0.0000	-3.09	0.0024 **	-	Yes
Precipitation	0.0010	1.00	0.3205	?	—
Temperature	-0.0375	-1.00	0.3204	-	Yes
ONI	-0.0070	-0.24	0.8121	-	Yes
COT net MM	0.0000	6.13	0.0000 ***	+	Yes

Table 8.2: MLR regression coefficients with expected vs observed signs — Robusta

Variable	Estimate	t-stat	p-value	Expected sign	Consistent
Sugar	1595.9308	2.25	0.0259 *	+	Yes
Cocoa	-0.2270	-3.08	0.0025 **	+	No
CPI	-20.7537	-3.11	0.0022 **	+	No
WTI	11.2848	5.96	0.0000 ***	+	Yes
Fed Funds Rate	46.0073	1.36	0.1754	-	No
DXY	15.3865	2.19	0.0304 *	-	No
Production	-0.0116	-1.46	0.1474	-	Yes
Consumption	0.0282	3.93	0.0001 ***	+	Yes
Exports	0.0083	0.83	0.4090	?	—
Ending stocks	0.0184	2.44	0.0160 *	-	No
Precipitation	1.4826	1.30	0.1974	?	—
Temperature	0.5158	0.01	0.9898	-	No
ONI	-49.5998	-1.46	0.1464	-	Yes
COT net MM	0.0035	3.65	0.0004 ***	+	Yes

Only 4 of the 14 predictors reach conventional significance ($p < 0.05$) for Arabica, against 8 for Robusta. For Arabica, COT net Managed Money is the strongest predictor ($t = 6.13$), followed by Sugar and WTI, both positive and significant; Ending stocks is negative and significant, consistent with inventory theory. The two remaining sign inconsistencies (CPI, Production) are both statistically insignificant; Exports carries no directional prior and is therefore excluded from this count. For Robusta, WTI, Sugar, Consumption, and COT net Managed Money are significant with the expected sign, but four further coefficients (CPI, DXY, Cocoa, Ending stocks) are significant with the wrong sign.

MLR regularisation: LASSO and Ridge. As a robustness extension to the OLS baseline, Ridge and LASSO regularised regression are also estimated under the identical walk-forward protocol. Table 8.3 compares the three specifications.

Table 8.3: MLR regularisation comparison — OLS vs LASSO vs Ridge

Specification	R ²	MSE	MAE
Arabica			
OLS (baseline)	0.7836	0.0352	0.1535
LASSO	0.7419	0.0420	0.1744
Ridge	0.6960	0.0494	0.1948
Robusta			
OLS (baseline)	0.2759	114285	274.6
LASSO	0.3849	97084	258.2
Ridge	0.4641	84583	247.5

Regularisation moves the two varieties in opposite directions. For Arabica, R^2 declines from 0.784 (OLS) to 0.742 (LASSO) and 0.696 (Ridge). For Robusta, R^2 rises from 0.276 (OLS) to 0.385 (LASSO) and 0.464 (Ridge), making Ridge the strongest linear specification tested for this variety. Across both varieties, LASSO shrinks only two of the 28 coefficients exactly to zero, CPI for Arabica and Temperature for Robusta, while Ridge leaves every coefficient in place but reduces its magnitude, with one exception: Arabica's ONI coefficient grows in magnitude, from -0.007 (OLS) to -0.024 (Ridge).

ARIMAX

Model order. Fixing the (p, d, q) order once on the initial training window, rather than re-selecting it at every walk-forward step, affects the two varieties asymmetrically, as Table 8.4 shows.

Table 8.4: ARIMAX order comparison — re-adaptive vs fixed order

Specification	Arabica R^2	Robusta R^2
Re-adaptive (order re-selected at every walk-forward step)	0.821	0.929
Fixed order (selected once, held fixed thereafter)	0.955	0.926

For Arabica, the fixed-order specification settles on ARIMA(2,0,1) and lifts R^2 from 0.821 to 0.955. For Robusta, it settles on ARIMA(1,0,3), and the change is marginal, from 0.929 to 0.926.

Regression coefficients. Table 8.5 and Table 8.6 report the coefficients of the 14 exogenous regressors from the reference model estimated on the full initial training window for each variety, including the t-statistics omitted from the simplified tables in Section 4.2.1.

Table 8.5: ARIMAX regression coefficients — Arabica

Variable	Estimate	t-stat	p-value
Sugar	1.0204	3.076	0.0021 **
Cocoa	0.0000	0.181	0.8565
CPI	0.0024	0.607	0.5435
WTI	0.0018	3.324	0.0009 ***
Fed Funds Rate	-0.0579	-4.566	0.0000 ***
DXY	-0.0023	-0.659	0.5098
Production	0.0000	-0.046	0.9636
Consumption	0.0000	0.495	0.6206

Predicting Arabica and Robusta Coffee Futures Prices

Variable	Estimate	t-stat	p-value
Exports	0.0000	−0.093	0.9256
Ending stocks	0.0000	0.151	0.8800
Precipitation	0.0001	0.503	0.6152
Temperature	0.0114	0.619	0.5360
ONI	0.0330	0.958	0.3381
COT net MM	0.0000	2.241	0.0250 *

Table 8.6: ARIMAX regression coefficients — Robusta

Variable	Estimate	t-stat	p-value
Sugar	317.4372	0.701	0.4832
Cocoa	−0.0128	−0.252	0.8011
CPI	14.1356	3.204	0.0014 **
WTI	2.5366	2.265	0.0235 *
Fed Funds Rate	35.6851	0.822	0.4112
DXY	−3.9373	−1.693	0.0905 .
Production	−0.0023	−0.497	0.6194
Consumption	−0.0008	−0.179	0.8581
Exports	−0.0023	−0.419	0.6749
Ending stocks	−0.0085	−1.487	0.1371
Precipitation	0.2623	0.921	0.3572
Temperature	−18.0120	−1.232	0.2179
ONI	−6.6544	−0.167	0.8677
COT net MM	0.0029	6.369	0.0000 ***

For Arabica, four regressors reach conventional significance: Sugar ($p = 0.0021$), WTI ($p < 0.001$), the Federal Funds Rate ($p < 0.001$, coefficient -0.058), and COT net Managed Money ($p = 0.025$). For Robusta, CPI ($p = 0.0014$), WTI ($p = 0.0235$), and COT net Managed Money ($p < 0.001$) are significant, and DX Y approaches significance at the 10% level ($p = 0.0905$).

Multicollinearity diagnostics. Table 8.7 reports the Variance Inflation Factor for each of the 14 regressors.

Table 8.7: Variance Inflation Factors — ARIMAX regressors

Variable	VIF (Arabica)	Diagnostic	VIF (Robusta)	Diagnostic
Sugar	4.21	Acceptable	4.21	Acceptable
Cocoa	3.17	Acceptable	2.55	Acceptable
CPI	36.68	Severe	27.12	Severe
WTI	6.94	Moderate	6.61	Moderate
Fed Funds Rate	3.27	Acceptable	2.50	Acceptable
DX Y	12.68	Severe	12.01	Severe
Production	42.16	Severe	36.22	Severe
Consumption	31.41	Severe	21.44	Severe
Exports	48.62	Severe	47.46	Severe
Ending stocks	4.17	Acceptable	4.34	Acceptable
Precipitation	1.71	Acceptable	1.78	Acceptable
Temperature	1.80	Acceptable	1.82	Acceptable
ONI	2.41	Acceptable	2.74	Acceptable
COT net MM	2.68	Acceptable	3.06	Acceptable

Predicting Arabica and Robusta Coffee Futures Prices

Exports, Production, Consumption, and CPI show severe multicollinearity (VIF above 20) for both varieties, while COT net Managed Money, the climate indices, Ending stocks, and the Federal Funds Rate all remain below 5.

Residual diagnostics. This matters because a well-specified model should leave residuals that resemble white noise: no remaining trend, autocorrelation, or systematic pattern the model failed to capture. Figure 8.1 shows the ACF and PACF of the reference model's residuals for both varieties. Table 8.8 confirms these residuals are stationary under both the ADF and KPSS tests.

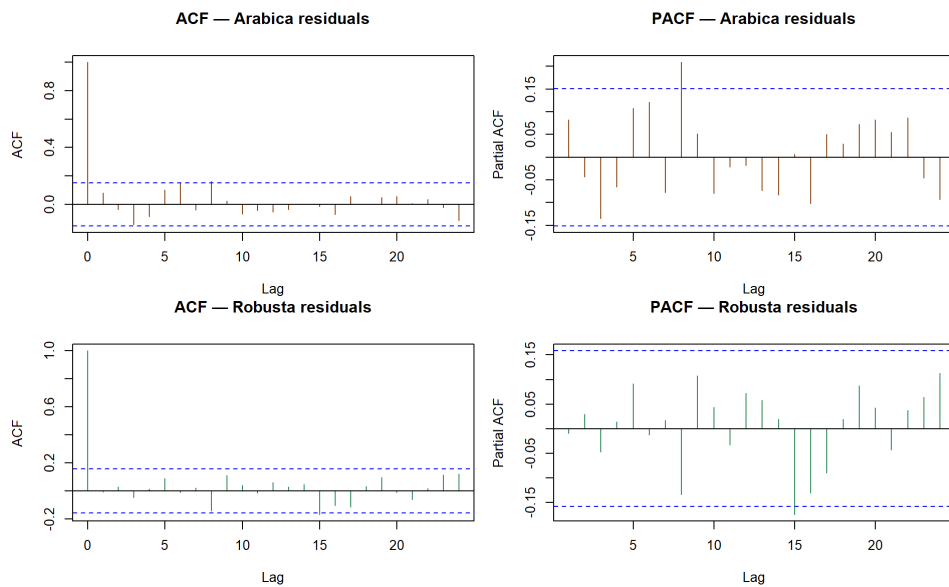


Figure 8.1: ACF and PACF of ARIMAX residuals — Arabica and Robusta

Table 8.8: ADF and KPSS stationarity tests — ARIMAX residuals

Series	ADF stat (p-val)	ADF result	KPSS stat (p-val)	KPSS result
ARIMAX residuals (Arabica)	-4.669 (p=0.01)	Stationary	0.334 (p=0.10)	Stationary
ARIMAX residuals (Robusta)	-4.579 (p=0.01)	Stationary	0.298 (p=0.10)	Stationary

8.3 Random Forest, XGBoost and SVM: Hyperparameter Sensitivity

This appendix reports the hyperparameter sensitivity analyses and tuning detail behind the ML summary in Section 4.1.1, for the three models whose full grids are not shown in the main text: Random Forest, XGBoost and SVM. The ANN, which involves more extensions, is reported separately.

Random Forest

The Random Forest model achieves an out-of-sample R^2 of 0.815 for Arabica and 0.676 for Robusta, averaging predictions across 500 regression trees, each grown on a bootstrap sample with a random subset of predictors considered at each split.

Tree count and complexity. Table 8.9 tests three levers of forest complexity separately, the number of trees, the minimum terminal node size, and the maximum number of terminal nodes, each varied while the other two are held at their default.

Table 8.9: Random Forest — tree count and complexity tuning

Group	Setting	R^2	MSE	MAE
Arabica				
Baseline	Default (500, 5, none)	0.8152	0.0082	0.0664
ntree	ntree = 100	0.8041	0.0087	0.0684
ntree	ntree = 300	0.8065	0.0086	0.0688
ntree	ntree = 1000	0.8031	0.0088	0.0691
nodesize	nodesize = 1	0.8235	0.0079	0.0668
nodesize	nodesize = 10	0.7788	0.0099	0.0733
nodesize	nodesize = 20	0.7146	0.0127	0.0846
maxnodes	maxnodes = 10	0.6987	0.0134	0.0861
maxnodes	maxnodes = 50	0.8038	0.0088	0.0692
maxnodes	maxnodes = 100	0.8042	0.0087	0.0694
Robusta				
Baseline	Default (500, 5, none)	0.6759	0.0214	0.1111
ntree	ntree = 100	0.6922	0.0204	0.1087
ntree	ntree = 300	0.6941	0.0202	0.1083
ntree	ntree = 1000	0.6961	0.0201	0.1078
nodesize	nodesize = 1	0.7324	0.0177	0.1020
nodesize	nodesize = 10	0.6497	0.0232	0.1164
nodesize	nodesize = 20	0.5869	0.0273	0.1250
maxnodes	maxnodes = 10	0.5929	0.0269	0.1257
maxnodes	maxnodes = 50	0.6945	0.0202	0.1083
maxnodes	maxnodes = 100	0.6935	0.0203	0.1079

The number of trees barely matters: Arabica stays within 0.803 to 0.807 and Robusta within 0.692 to 0.696 across the full 100-to-1000 range. Node size and maximum nodes matter considerably more, and in the same direction for both varieties: allowing trees to grow more freely, through a smaller nodesize or a higher maxnodes, improves performance, while constraining them too aggressively hurts it. The single largest gain comes from nodesize = 1 for Robusta, lifting R^2 from 0.676 to 0.732; the single largest loss comes from maxnodes = 10 for Arabica, dropping R^2 to 0.699.

Comparison with a single pruned tree. Table 8.10 compares the Random Forest baseline against a single CART tree, pruned via the one-standard-error rule on its cost-complexity path.

Table 8.10: Random Forest vs a single pruned CART tree

Specification	R ²	MSE	MAE
Arabica			
Random Forest (baseline)	0.8152	0.0082	0.0664
Single pruned tree (CART, cp via 1-SE rule)	0.6326	0.0164	0.1021
Robusta			
Random Forest (baseline)	0.6759	0.0214	0.1111
Single pruned tree (CART, cp via 1-SE rule)	0.3148	0.0453	0.1624

A single pruned tree falls well short of the forest for both varieties, losing 0.183 of R² for Arabica and 0.361 for Robusta. Averaging over many bootstrap-grown trees cancels out noise that no amount of pruning can replicate within a single tree.

Cost-complexity pruning via scikit-learn. R's randomForest implementation has no cp argument, since cost-complexity pruning belongs to a single CART tree rather than an ensemble. Table 8.11 tests scikit-learn's RandomForestRegressor, which does expose a ccp_alpha parameter, applying it to every tree in the forest³.

Table 8.11: Random Forest — R baseline vs scikit-learn with cost-complexity pruning

Specification	R ²	MSE	MAE
Arabica			
randomForest (R), unpruned baseline	0.8152	0.0082	0.0664
scikit-learn RandomForestRegressor, ccp_alpha = 0 (unpruned)	0.8089	0.0085	0.0692
scikit-learn RandomForestRegressor, ccp_alpha = 0.001	0.6381	0.0162	0.0982
scikit-learn RandomForestRegressor, ccp_alpha = 0.01	0.0854	0.0408	0.1666
Robusta			
randomForest (R), unpruned baseline	0.6759	0.0214	0.1111
scikit-learn RandomForestRegressor, ccp_alpha = 0 (unpruned)	0.6798	0.0212	0.1123
scikit-learn RandomForestRegressor, ccp_alpha = 0.001	0.4853	0.0340	0.1463
scikit-learn RandomForestRegressor, ccp_alpha = 0.01	-0.0771	0.0712	0.2138

The unpruned scikit-learn specification (ccp_alpha = 0) closely reproduces the R baseline for both varieties, confirming that cost-complexity pruning is not fundamentally incompatible with a Random Forest. But applying it at all is actively harmful: even a mild

³ <https://scikit-learn.org/stable/modules/tree.html#minimal-cost-complexity-pruning>

penalty ($ccp_alpha = 0.001$) collapses R^2 to 0.638 for Arabica and 0.485 for Robusta, and a moderate one ($ccp_alpha = 0.01$) drives it to 0.085 and -0.077 , a negative value indicating that the model performs worse than simply predicting the mean of the test period at every step. Pruning individual trees removes the diversity across trees that the ensemble's averaging depends on, without addressing an overfitting risk the ensemble did not have to begin with.

XGBoost

XGBoost is tuned via a three-split time series cross-validation search (Section 3.5.7), selecting 90 boosting rounds, a maximum depth of 4, a learning rate of 0.08, and a subsampling fraction of 0.8 for Arabica, against 100 rounds, a maximum depth of 3, a learning rate of 0.3, and a subsampling fraction of 0.9 for Robusta. Table 8.12 reports the resulting out-of-sample performance.

Table 8.12: XGBoost — tuned hyperparameters and performance by variety

Series	nrounds	max_depth	eta	subsample	R^2	MSE	MAE
Arabica	90	4	0.08	0.8	0.8117	0.0084	0.0676
Robusta	100	3	0.30	0.9	0.7970	0.0134	0.0874

XGBoost reaches an R^2 of 0.812 for Arabica and 0.797 for Robusta, the strongest of the four machine learning models for Robusta specifically. The two varieties converge on quite different configurations: Arabica favours a slower, more conservative learning process, while Robusta favours a faster, shallower one, consistent with its shorter and more volatile training sample.

SVM

The Support Vector Machine is tuned via the same three-split time series cross-validation procedure. For Robusta, this search improves on the original specification, selecting $cost = 10$ and $gamma = 0.05$; for Arabica, every tuned combination underperforms the original specification ($cost = 5$, $gamma = 0.05$) on the test set, so these values are retained. Table 8.13 reports the resulting performance.

Table 8.13: SVM — hyperparameters and performance by variety

Series	cost	gamma	R ²	MSE	MAE
Arabica	5	0.05	0.8130	0.0083	0.0564
Robusta	10	0.05	0.7607	0.0158	0.0995

SVM reaches an R² of 0.813 for Arabica, identical to the untuned baseline, and 0.761 for Robusta, an improvement over the baseline of 0.723. The higher cost selected for Robusta allows a tighter margin, letting the model fit the shorter, more volatile series more closely without an apparent loss of generalisation.

8.4 Artificial Neural Network: Extensions and Sensitivity Analysis

The tuned ANN, selected via three-split time series cross-validation over neuron count and L2 regularisation strength (Section 3.5.9), reaches an R² of 0.728 for Arabica and 0.456 for Robusta, an improvement over the untuned baseline of 0.453 and 0.441 respectively. Both varieties converge on the smallest architecture tested, two hidden neurons, with a decay of 0.1 for Arabica and 0.001 for Robusta. This appendix reports the extensions behind the ANN summary in Section 4.1.1: the software used, and its sensitivity to neuron count, network depth, and activation function.

Software implementation. Table 8.14 summarises the four R packages used across the extensions in this section, since each supports a different combination of network depth, regularisation, and optimiser, and no single package covers all three.

Table 8.14: R packages used for the ANN extensions in this study — capabilities and role

Package	Multi-layer depth	L2 regularisation	Optimiser	Role in this study
nnet	No (single layer only)	Yes (decay)	Quasi-Newton (BFGS)	Main tuned ANN specification (reported throughout Chapter 4)
neuralnet	Yes	No	Resilient backpropagation (Rprop+)	Isolates the effect of depth alone (Table 8.16, undecayed rows)
RSNNS	Yes	Yes (via BackpropWeightDecay)	Fixed-step gradient descent	Depth with decay, but under-optimised relative to nnet/neuralnet (Table 8.16, decayed rows)
keras3	No (single layer tested)	Yes	Adam, with early stopping	Activation function × L2 regularisation (Table 8.18), the most robust optimiser of the four

Sensitivity to neuron count. Table 8.15 isolates the effect of neuron count alone, holding decay fixed at each variety's selected value.

Table 8.15: ANN — validation MSE by number of hidden neurons

Neurons	Validation MSE
Arabica	
2	0.0353
3	0.0356
5	0.0358
8	0.0358
10	0.0359
15	0.0359
Robusta	
2	0.0450
3	0.1832
5	0.0528
8	0.0815
10	0.0678
15	0.0761

For Arabica, validation error is essentially flat across neuron counts, rising only slightly from 0.0353 at 2 neurons to 0.0359 at 15. Robusta behaves quite differently: validation error jumps more than fourfold between 2 neurons (0.0450) and 3 neurons (0.1832), a pattern more consistent with an isolated poor local optimum reached during training than a genuine relationship between network width and performance, given that nnet's random weight initialisation offers no guarantee of finding the global minimum on a non-convex loss surface.

Network depth. Table 8.16 tests deeper architectures, two, five and ten hidden layers, first without weight decay (neuralnet, which has no built-in regularisation mechanism), then with decay added via RSNNS, the only package tested that supports both multiple layers and weight decay simultaneously.

Table 8.16: ANN — depth and weight decay comparison

Architecture	R ²	MSE	MAE
Arabica			
1 layer (baseline, nnet)	0.7282	0.0121	0.0924
2 layers	0.5981	0.0179	0.1126
5 layers	0.3049	0.0310	0.1376
10 layers	-0.3931	0.0622	0.2037
2 layers + decay	-1.2055	0.0984	0.2578
5 layers + decay	-1.0537	0.0917	0.2510
10 layers + decay	-1.2199	0.0991	0.2560
Robusta			
1 layer (baseline, nnet)	0.4560	0.0360	0.1491

Architecture	R ²	MSE	MAE
2 layers	0.3351	0.0440	0.1691
5 layers	−0.1281	0.0746	0.2311
10 layers	−0.2751	0.0843	0.2573
2 layers + decay	−0.4227	0.0941	0.2676
5 layers + decay	−0.2641	0.0836	0.2518
10 layers + decay	−0.3669	0.0904	0.2655

Depth alone is actively harmful: R^2 falls monotonically as layers are added, turning negative by five layers for Robusta (−0.128) and by ten layers for Arabica (−0.393). Adding weight decay at depth makes this worse rather than better; every decayed configuration underperforms its undecayed counterpart. This most likely reflects a methodological confound rather than a genuine effect of regularisation: accessing decay in RSNNs requires switching to a slower, less efficient optimiser (BackpropWeightDecay) than the one used elsewhere (Rprop+ in neuralnet, Adam in Keras), leaving the deeper decayed networks under-optimised within the fixed iteration budget rather than genuinely worse off from the regularisation itself.

Activation function. Table 8.17 compares sigmoid and tanh activation within neuralnet, at the single-layer, two-neuron architecture, without regularisation.

Table 8.17: ANN — sigmoid vs tanh activation function (neuralnet, unregularised)

Activation	R ²	MSE	MAE
Arabica			
Sigmoid (logistic)	0.5774	0.0189	0.1080
Tanh	0.6176	0.0171	0.1052
Robusta			
Sigmoid (logistic)	0.3339	0.0441	0.1699
Tanh	0.2881	0.0471	0.1665

Tanh outperforms sigmoid for Arabica ($R^2 = 0.618$ vs 0.577) and underperforms it for Robusta (0.288 vs 0.334), a modest but real difference running in opposite directions for the two varieties.

Activation function with L2 regularisation. Table 8.18 repeats this comparison using Keras, which supports both a configurable activation function and L2 regularisation on the same layer, testing all four combinations.

Table 8.18: ANN (Keras) — activation function $\times$ L2 regularisation

Configuration	R ²	MSE	MAE
Arabica			
Sigmoid, no L2	−0.2168	0.0543	0.1682
Sigmoid, L2	−0.2348	0.0551	0.1887
Tanh, no L2	0.7124	0.0128	0.0942
Tanh, L2	0.3944	0.0270	0.1292
Robusta			
Sigmoid, no L2	0.2227	0.0514	0.1950
Sigmoid, L2	0.2127	0.0521	0.1982
Tanh, no L2	0.5215	0.0316	0.1470
Tanh, L2	0.4869	0.0339	0.1514

The ranking reverses: Keras's Adam optimiser finds tanh consistently stronger than sigmoid for both varieties (Arabica: 0.712 vs −0.217; Robusta: 0.522 vs 0.223), with sigmoid even underperforming a naive mean-prediction baseline for Arabica specifically. L2 regularisation uniformly hurts performance rather than helping. This may reflect Keras's early stopping already preventing the network from overfitting on its own, so that an extra penalty on top only makes the model more cautious than necessary, without any further benefit. The disagreement between neuralnet and Keras on which activation performs better for Robusta shows that the size, and even the direction, of this effect depends on the optimiser used to fit the network, not on the activation function alone.

8.5 Variable Importance and Partial Dependence: Full Detail

Section 4.2.2 discusses the top three predictors by permutation importance for the four machine learning models, and the partial dependence of those consensus drivers. This appendix reports the complete detail behind that discussion: the full ranking of all 14 predictors for all six models, including MLR and ARIMAX, and the partial dependence plots for every predictor shown in Section 4.2.2, for all four machine learning models.

Permutation Importance — Full Rankings

As already noted in Section 4.2.2, MLR and ARIMAX are excluded from the comparative chart in the main text: MLR because severe multicollinearity produces negative importance scores throughout, and ARIMAX because it predicts on the original price scale

rather than the Min-Max scaled range used by the four ML models. The full rankings below include both, with this caveat in mind. For MLR, both the model and the test set used here are Min-Max scaled, so the resulting Δ MSE values are internally consistent but on a different basis than the original-price-unit metrics reported in Table 4.1. For ARIMAX, the reference model is calibrated on the original price scale but is evaluated here against a Min-Max scaled test set, a mismatch between the model and its inputs; the resulting ranking should therefore be read as indicative rather than with the same confidence as the results reported in the main text.

MLR. Table 8.19 reports the full ranking. Every score is negative for both varieties, and for Robusta every predictor returns an almost identical, extremely large negative score.

Table 8.19: Permutation importance — MLR, full ranking

Variable	Δ MSE
Arabica	
WTI	-1.8162
COT net MM	-1.8213
Ending stocks	-1.8235
Sugar	-1.8251
Consumption	-1.8324
Temperature	-1.8330
Precipitation	-1.8334
ONI	-1.8340
Fed Funds Rate	-1.8344
Cocoa	-1.8345
CPI	-1.8349
Exports	-1.8354
Production	-1.8355
DXY	-1.8362
Robusta	
WTI	-3,545,583
Fed Funds Rate	-3,545,583
DXY	-3,545,583
Sugar	-3,545,583
COT net MM	-3,545,583
Exports	-3,545,583
Production	-3,545,583
Temperature	-3,545,583
Precipitation	-3,545,583
Consumption	-3,545,583
ONI	-3,545,583
Ending stocks	-3,545,583
Cocoa	-3,545,583
CPI	-3,545,583

ARIMAX. Table 8.20 reports the full ranking, in the original price units noted above.

Sugar and the Federal Funds Rate stand out from the rest for Arabica; for Robusta, Sugar and

the Federal Funds Rate lead again, though the gap between the top two and the remaining twelve is, proportionally, much narrower than for Arabica.

Table 8.20: Permutation importance — ARIMAX, full ranking

Variable	AMSE
Arabica	
Sugar	1.8054
Fed Funds Rate	1.7769
WTI	1.7689
DXY	1.7688
COT net MM	1.7688
Consumption	1.7688
Exports	1.7688
Production	1.7688
Ending stocks	1.7688
Precipitation	1.7688
Cocoa	1.7688
CPI	1.7688
Temperature	1.7682
ONI	1.7668
Robusta	
Sugar	4,294,599
Fed Funds Rate	4,278,189
CPI	4,275,898
WTI	4,275,428
Precipitation	4,275,390
Ending stocks	4,275,387
COT net MM	4,275,386
Consumption	4,275,386
Production	4,275,386
Exports	4,275,386
Cocoa	4,275,385
DXY	4,275,305
ONI	4,274,947
Temperature	4,274,362

Random Forest. Table 8.21 reports the full ranking behind the top three already discussed in Section 4.2.2.

Table 8.21: Permutation importance — Random Forest, full ranking

Variable	AMSE
Arabica	
Ending stocks	0.0445
COT net MM	0.0244
Fed Funds Rate	0.0234
ONI	0.0202
WTI	0.0199
Sugar	0.0194
Production	0.0188
Exports	0.0184
Consumption	0.0181
Precipitation	0.0181
CPI	0.0181
DXY	0.0180
Temperature	0.0177
Cocoa	0.0165

Predicting Arabica and Robusta Coffee Futures Prices

Variable	Δ MSE
Robusta	
WTI	0.1299
Fed Funds Rate	0.1242
Sugar	0.1232
COT net MM	0.1230
Ending stocks	0.1227
Cocoa	0.1223
Exports	0.1221
CPI	0.1219
Temperature	0.1218
Precipitation	0.1216
Production	0.1208
DXY	0.1205
Consumption	0.1202
ONI	0.1189

XGBoost. Table 8.22 reports the full ranking behind the top three already discussed in Section 4.2.2.

Table 8.22: Permutation importance — XGBoost, full ranking

Variable	Δ MSE
Arabica	
Ending stocks	0.0585
COT net MM	0.0333
WTI	0.0250
ONI	0.0238
Precipitation	0.0225
Fed Funds Rate	0.0223
Consumption	0.0222
Production	0.0222
CPI	0.0221
Exports	0.0221
DXY	0.0220
Temperature	0.0217
Cocoa	0.0200
Sugar	0.0191
Robusta	
Fed Funds Rate	0.1084
DXY	0.1072
Sugar	0.1071
WTI	0.1047
Ending stocks	0.1030
Temperature	0.1020
Consumption	0.1020
Production	0.1018
Exports	0.1018
CPI	0.1018
Precipitation	0.1014
COT net MM	0.1010
Cocoa	0.1009
ONI	0.0985

SVM. Table 8.23 reports the full ranking behind the top three already discussed in Section 4.2.2. For Arabica, the gap between the top predictor and the fourteenth is

comparatively narrow relative to the other three ML models; for Robusta, all four ML models show a similarly narrow spread, unlike the near-uniform collapse seen for MLR.

Table 8.23: Permutation importance — SVM, full ranking

Variable	AMSE
Arabica	
WTI	0.0765
Ending stocks	0.0724
COT net MM	0.0696
Sugar	0.0675
Fed Funds Rate	0.0666
Exports	0.0661
Production	0.0655
Precipitation	0.0653
ONI	0.0648
Temperature	0.0641
Cocoa	0.0641
Consumption	0.0628
CPI	0.0576
DXY	0.0565
Robusta	
WTI	0.1525
Fed Funds Rate	0.1503
COT net MM	0.1468
Sugar	0.1421
Ending stocks	0.1421
Cocoa	0.1419
Precipitation	0.1412
Production	0.1402
ONI	0.1393
Consumption	0.1392
Exports	0.1388
Temperature	0.1375
DXY	0.1345
CPI	0.1328

ANN. Table 8.24 reports the full ranking behind the top three already discussed in Section 4.2.2. The gap between the top four predictors and the remaining ten is considerably wider for Arabica than for Robusta, where all 14 scores stay within a comparatively narrow band.

Table 8.24: Permutation importance — ANN, full ranking

Variable	AMSE
Arabica	
Ending stocks	0.0172
Sugar	0.0153
WTI	0.0153
COT net MM	0.0143
ONI	0.0081
CPI	0.0065
Fed Funds Rate	0.0064
Temperature	0.0062
Consumption	0.0056

Predicting Arabica and Robusta Coffee Futures Prices

Variable	Δ MSE
Precipitation	0.0051
Cocoa	0.0048
Production	0.0041
Exports	0.0039
DXY	0.0008
Robusta	
WTI	0.4815
Exports	0.4577
Fed Funds Rate	0.4569
Sugar	0.4569
Ending stocks	0.4569
Production	0.4547
Temperature	0.4541
COT net MM	0.4529
Consumption	0.4488
ONI	0.4488
DXY	0.4487
Precipitation	0.4471
Cocoa	0.4341
CPI	0.3533

Partial Dependence Plots — Per-Model Detail

Figures 1 through 8 report the partial dependence plots behind the discussion in Section 4.2.2, for all four machine learning models and both varieties.

Random Forest. For Arabica, COT net Managed Money rises smoothly before flattening; Ending stocks drops sharply at a single threshold before flattening, with a slight uptick at the extreme upper end; and the predicted price falls as the Fed Funds Rate rises, flattening past a rate of roughly 2%. For Robusta, the Fed Funds Rate and Sugar both rise gently before plateauing, while WTI stays flat until a sharp jump around 80 USD/barrel, after which it plateaus again.

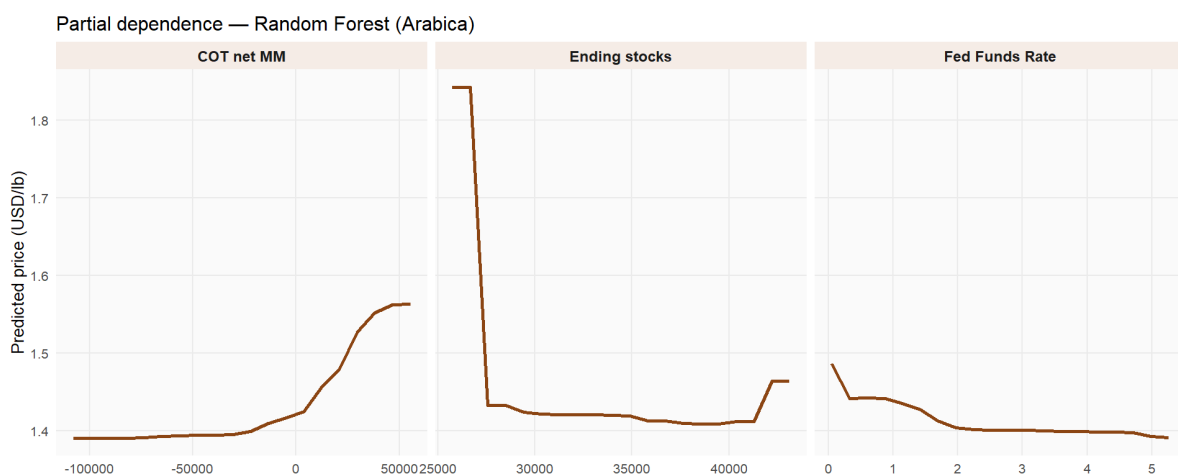


Figure 8.2: Partial dependence — Random Forest (Arabica)

Predicting Arabica and Robusta Coffee Futures Prices

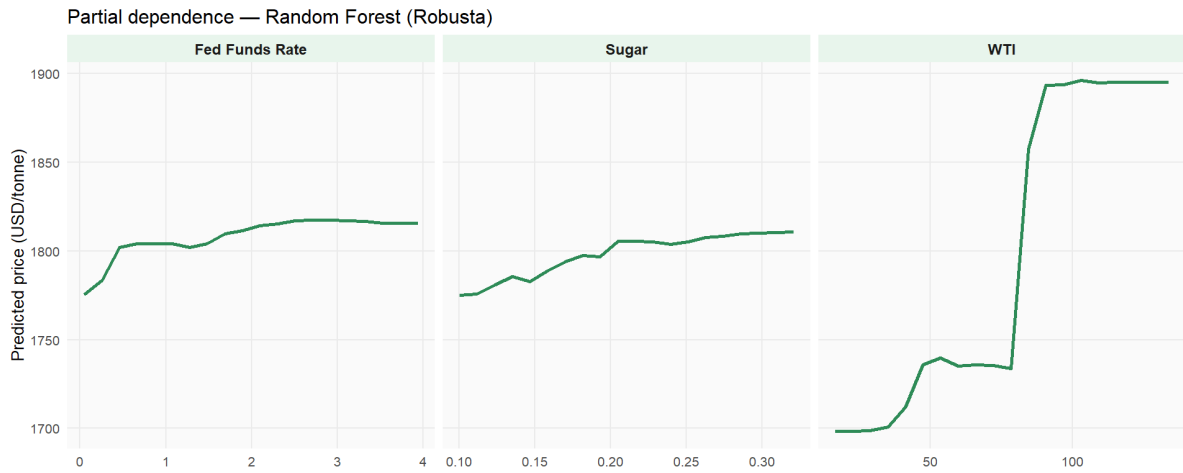


Figure 8.3: Partial dependence — Random Forest (Robusta)

XGBoost. The Arabica plots share two of Random Forest’s three variables, Ending stocks and COT net Managed Money, in a closely similar shape; WTI replaces the Fed Funds Rate as the third driver, rising steadily across its range. For Robusta, DXY traces a U-shape, falling sharply before recovering; Sugar moves in a choppy but broadly rising pattern; and the Fed Funds Rate rises gently before flattening, similarly to Random Forest.

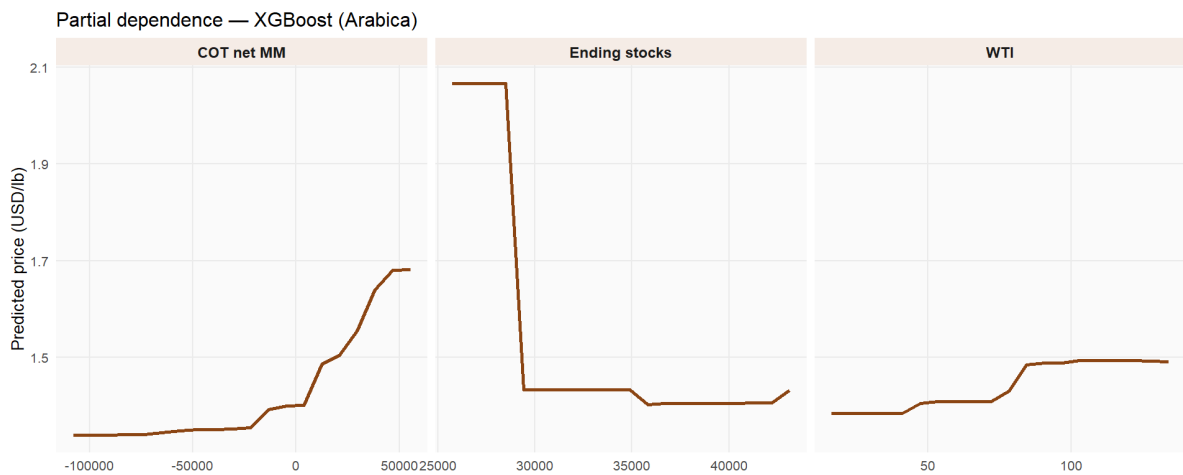


Figure 8.4: Partial dependence — XGBoost (Arabica)

Predicting Arabica and Robusta Coffee Futures Prices

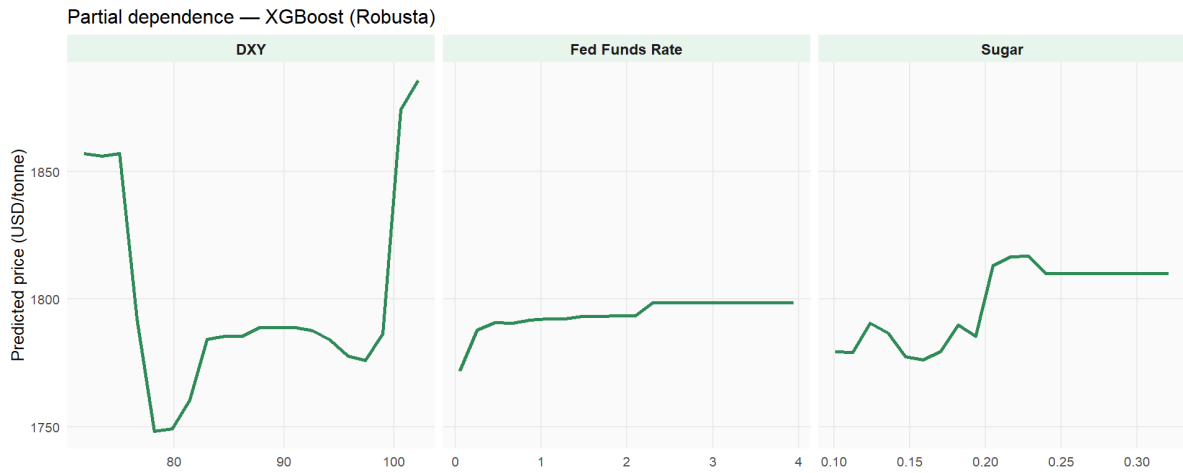


Figure 8.5: Partial dependence — XGBoost (Robusta)

SVM. Unlike the two tree-based models, SVM produces smooth, near-monotonic curves throughout. For Arabica, COT net Managed Money and WTI rise steadily while Ending stocks declines steadily. For Robusta, COT net Managed Money and WTI again rise smoothly, while the Fed Funds Rate rises before turning down slightly past a rate of roughly 3%.

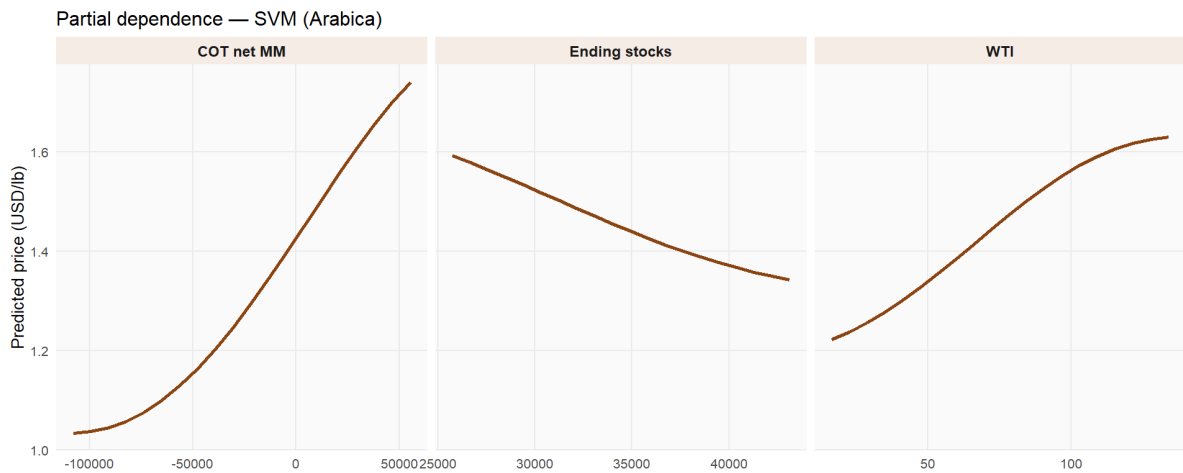


Figure 8.6: Partial dependence — SVM (Arabica)

Predicting Arabica and Robusta Coffee Futures Prices

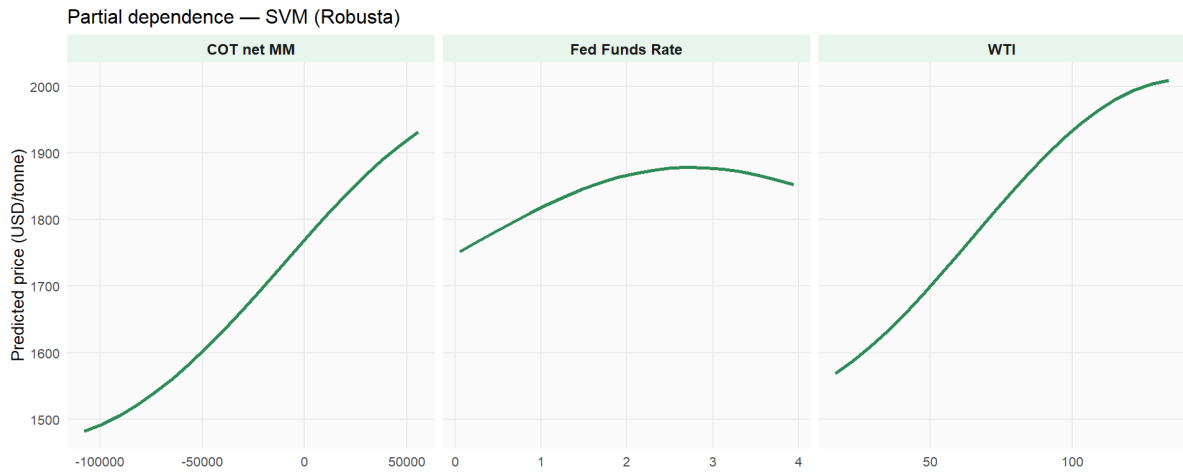


Figure 8.7: Partial dependence — SVM (Robusta)

ANN. For Arabica, Ending stocks declines while Sugar and WTI rise, all close to linearly. For Robusta, Exports declines steadily, while the Fed Funds Rate and WTI both rise, the latter particularly steeply.

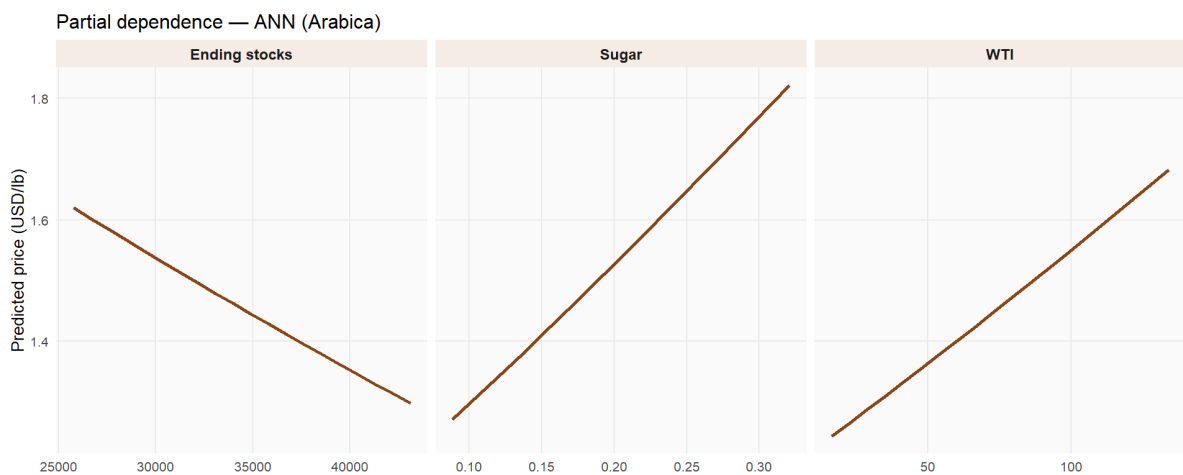


Figure 8.8: Partial dependence — ANN (Arabica)

Predicting Arabica and Robusta Coffee Futures Prices

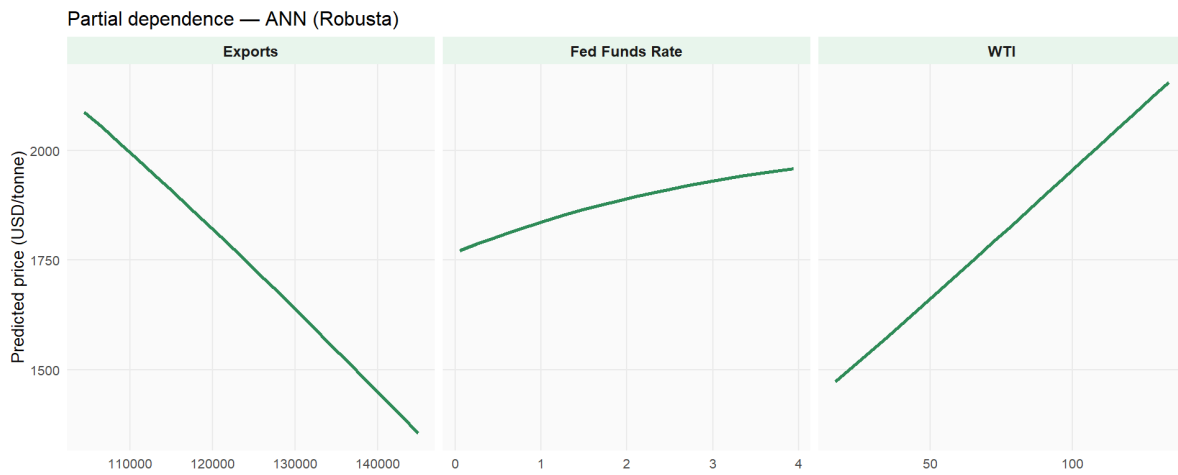


Figure 8.9: Partial dependence — ANN (Robusta)

8.6 Robustness Checks

Four robustness checks test the sensitivity of the main results to specific methodological choices: the aggregation of climate variables across producer countries, the use of global rather than variety-specific climate and supply-and-demand data, the monthly estimation frequency, and the treatment of speculative positioning as a contemporaneous rather than lagged signal.

8.6.1 Disaggregated Climate Variables

The main specifications rely on two production-weighted climate indices that aggregate temperature and precipitation conditions across 14 producer countries into single monthly series. To assess whether relevant country-specific climate information is lost through this aggregation, the four ML models are re-estimated under an alternative specification in which the two aggregated indices are replaced by the 28 individual national series (14 countries multiplied by 2 climate variables), holding all other predictors constant and yielding 40 regressors in total. A comparison of out-of-sample R^2 between the two configurations provides direct evidence on whether disaggregation improves predictive accuracy. ARIMAX and MLR are excluded from this comparison: the former would require re-estimating the automatic ARIMA selection procedure with 40 exogenous regressors at every step of the walk-forward loop, which is computationally prohibitive, and the latter is ill-conditioned with 40 predictors on approximately 170 training observations.

Table 8.25 compares the four ML models' out-of-sample R^2 under the aggregated (baseline) and disaggregated (40-regressor) climate specifications.

Table 8.25: Robustness check — aggregated vs disaggregated climate indices

Model	R^2 (aggregated)	R^2 (disaggregated)	ΔR^2
Arabica			
Random Forest	0.8152	0.7632	-0.0520
XGBoost	0.8117	0.7977	-0.0140
SVM	0.8130	-0.0137	-0.8267
ANN	0.7282	-0.1864	-0.9146
Robusta			
Random Forest	0.6759	0.5790	-0.0969
XGBoost	0.7970	0.7384	-0.0586
SVM	0.7607	-0.1064	-0.8671
ANN	0.4560	-0.3279	-0.7839

Disaggregation deteriorates performance for every model-variety combination, without exception. The decline is moderate for Random Forest and XGBoost, but severe for SVM and ANN, both of which turn negative for both varieties. This is consistent with the curse of dimensionality: with 40 regressors and roughly 170 training observations, models more sensitive to input dimensionality overfit and fail to generalise. The result validates the production-weighted aggregation used throughout the main specifications: aggregating filters out country-specific noise that the disaggregated series cannot compensate for at this sample size.

8.6.2 Variety-Specific Model Specification

The second robustness check re-estimates all six models under a variety-specific specification in which global climate indices and world supply-and-demand fundamentals are replaced by their variety-relevant counterparts. For Arabica models, climate indices are aggregated exclusively across the twelve Arabica-producing countries using variety-adjusted production weights, and the four supply-demand variables are constructed by summing country-level values weighted by empirically computed Arabica variety split ratios. The same procedure is applied symmetrically for Robusta models, using the six Robusta-producing countries and their corresponding variety splits. All other predictors, including macroeconomic

variables, the speculative positioning variable and the ONI index, are retained unchanged, as no variety-specific equivalents are available or meaningful for these series. This specification tests whether restricting climate and supply information to the actual producing zones for each variety improves out-of-sample accuracy relative to the global baseline.

Table 8.26 compares all six models' out-of-sample R^2 under the baseline (global) and variety-specific specifications.

Table 8.26: Baseline vs variety-specific specification

Model	R^2 (baseline)	R^2 (variety-specific)	ΔR^2
Arabica			
MLR	0.7836	0.5517	-0.2319
ARIMAX	0.9547	0.5517	-0.4030
Random Forest	0.8152	0.7329	-0.0823
XGBoost	0.8117	0.7160	-0.0957
SVM	0.8130	0.6846	-0.1284
ANN	0.7282	-0.4130	-1.1412
Robusta			
MLR	0.2759	0.2740	-0.0019
ARIMAX	0.9257	0.9360	+0.0103
Random Forest	0.6759	0.7205	+0.0446
XGBoost	0.7970	0.8489	+0.0519
SVM	0.7607	0.6093	-0.1514
ANN	0.4560	0.4934	+0.0374

The two varieties respond in opposite directions. For Robusta, four of six models improve, most notably XGBoost, which reaches $R^2 = 0.849$, the strongest result obtained anywhere in this study. For Arabica, every model deteriorates. ARIMAX is the most striking case: under the variety-specific specification, the automatic order selection collapses to ARIMA(0,0,0), meaning no autoregressive or moving-average term improves on a pure regression on the exogenous predictors, so ARIMAX becomes numerically identical to MLR ($R^2 = 0.552$ for both), a fall of 0.403 from its monthly baseline of 0.955. The ANN falls furthest of all six models under this specification, to a negative R^2 of -1.141 for Arabica.

This asymmetry is plausibly linked to the two varieties' different production geographies. Robusta production is concentrated in a handful of countries, Vietnam and Brazil alone account for over two-thirds of the weighted aggregate, so restricting both climate and

fundamentals to just those countries loses little of the original global signal, and four of six models hold steady or improve as a result. Arabica production is more geographically diversified, so the same restriction barely changes the climate signal given Brazil's already-dominant weight, but it narrows the fundamentals variables (production, consumption, exports, ending stocks) from a world total to a twelve-country total. Every one of the six Arabica models deteriorates under this narrower specification, two of them severely, suggesting the fundamentals variables lose real information for Arabica when restricted to just twelve countries.

8.6.3 OECD GDP Growth (Quarterly Specification)

A third robustness check addresses a variable suggested during supervision: OECD real GDP growth, a demand-side macroeconomic indicator published only at quarterly frequency, with no monthly equivalent available. Incorporating it into the main monthly pipeline is therefore not possible without an artificial frequency conversion, and a quarterly re-estimation is required instead. Because changing the estimation frequency and adding a predictor are two distinct methodological changes, three configurations are compared rather than two. The first is the monthly baseline already estimated in the main specification. The second aggregates the same fourteen baseline predictors to quarterly frequency by averaging the three constituent months, without adding GDP growth, isolating the effect of the frequency change alone. The third adds OECD GDP growth as a fifteenth predictor to the quarterly specification. Comparing the second and third configurations isolates the marginal contribution of GDP growth at a fixed quarterly frequency, while comparing the first and second configurations indicates the cost, in terms of out-of-sample accuracy, of reducing the available sample to quarterly observations. All six models are re-estimated under each quarterly configuration using the same walk-forward expanding-window protocol and 80/20 split convention as the monthly baseline.

Table 8.27 compares three configurations for all six models: the monthly baseline, a quarterly specification without GDP growth (isolating the effect of the frequency change alone), and a quarterly specification with GDP growth added as a fifteenth predictor.

Table 8.27: Monthly baseline vs quarterly specifications, without and with GDP growth

Model	R ² (monthly)	R ² (quarterly, no GDP)	R ² (quarterly, with GDP)	ΔR ²
Arabica				
MLR	0.7836	0.8201	0.7778	−0.0423
ARIMAX	0.9547	0.7413	0.7778	+0.0365
Random Forest	0.8152	0.6658	0.6699	+0.0041
XGBoost	0.8117	0.7445	0.7179	−0.0266
SVM	0.8130	0.4459	0.3228	−0.1231
ANN	0.7282	−0.9511	−1.7509	−0.7998
Robusta				
MLR	0.2759	−0.0544	−0.0504	+0.0040
ARIMAX	0.9257	0.3613	0.6156	+0.2543
Random Forest	0.6759	0.2453	0.2408	−0.0045
XGBoost	0.7970	0.4432	0.4513	+0.0081
SVM	0.7607	0.2903	0.0666	−0.2237
ANN	0.4560	−0.0753	0.1260	+0.2013

The frequency change dominates. Moving from monthly to quarterly data deteriorates most models, most severely the ANN, which collapses to a strongly negative R² for Arabica (−0.951) even before GDP growth is added. ARIMAX is affected in a specific way: at quarterly frequency, the automatic order search again settles on ARIMA(0,0,0), so the model reduces to a static regression identical to MLR, both reaching R² = 0.778 for Arabica once GDP growth is included, echoing the same collapse observed in the variety-specific check described above. The most plausible explanation for the machine learning models is sample size: the quarterly training window covers only 56 observations for Arabica and 51 for Robusta, against 168 and 153 at monthly frequency, disproportionately penalising the more data-hungry specifications.

Once frequency is held fixed, GDP growth itself adds little. For Arabica, only ARIMAX and Random Forest improve, and both marginally; for Robusta, the gains are somewhat larger, most notably for ARIMAX (+0.254) and ANN (+0.201), though both remain well below their monthly baselines. Overall, the check provides no support for including GDP growth as a predictor, and the frequency cost of testing it remains the dominant finding.

8.6.4 Lagged COT Net Managed Money Position

The discussion of the Variable Importance results in Chapter 4 raised the possibility that the COT net Managed Money position is involved in a self-reinforcing dynamic with prices, where positioning and price changes feed into each other over time rather than COT simply acting as a contemporaneous signal. If that is genuinely happening, then past values of COT net MM, not just the current month's value, should carry some additional predictive content. This robustness check tests that directly by adding the COT position lagged by one, two and three months to the baseline predictor set, alongside the contemporaneous COT variable already used throughout the study. The comparison is not between COT and no COT, but between the baseline 14-predictor specification and an extended 17-predictor specification, which isolates the marginal contribution of the lagged values specifically. All four ML models are re-estimated for both Arabica and Robusta under the extended specification, following the same walk-forward protocol used in the main results.

Table 8.28 compares the baseline 14-predictor specification against an extended 17-predictor specification that adds the COT net Managed Money position lagged by one, two and three months, for the four ML models.

Table 8.28: Robustness check — baseline vs lagged COT specification

Model	R ² (baseline)	R ² (with COT lags)	ΔR^2
Arabica			
Random Forest	0.8152	0.8060	-0.0092
XGBoost	0.8117	0.8197	+0.0080
SVM	0.8130	0.7208	-0.0922
ANN	0.7282	0.1837	-0.5445
Robusta			
Random Forest	0.6759	0.6623	-0.0136
XGBoost	0.7970	0.7433	-0.0537
SVM	0.7607	0.6656	-0.0951
ANN	0.4560	0.5269	+0.0709

Six of the eight model-variety combinations deteriorate under the extended specification; only XGBoost for Arabica (+0.008) and the ANN for Robusta (+0.071) improve, both modestly. The clearest deterioration is the Arabica ANN, which collapses from 0.728 to

0.184, a loss of 0.545. This is best explained by redundancy rather than dimensionality as such: the three lagged COT values are, by construction, highly autocorrelated with the contemporaneous position already in the baseline specification, so they add little new information while still requiring the model to estimate three additional parameters on the same limited training sample. The check provides little support for the idea that recent COT history carries additional predictive content beyond what the contemporaneous position already captures. This is consistent with Ederer et al. (2016), who find that the price response to a shock in money managers' net long positions dissipates within a single month, a result they attribute to the speed at which financial markets process new information.